

Sishen pills inhibit inflammatory dendritic cell differentiation via miR-505–3p mediated E-cadherin downregulation in ulcerative colitis

Jiaqi Huang^a, Youbao Zhong^b, Nian Cheng^a, Zheyang Zhang^a, Li Huang^a, Lizhao Song^a, Shaomin Cheng^{c,d,*}, Haimei Zhao^{c,d,**}, Duanyong Liu^{c,e,**}

^a Department of Postgraduate, Jiangxi University of Chinese Medicine, Nanchang 330004, Jiangxi Province, China

^b Laboratory Animal Research Center for Science and Technology, Jiangxi University of Chinese Medicine, Nanchang 330004, Jiangxi Province, China

^c Formula-Pattern Research Center of Jiangxi University of Chinese Medicine, Nanchang 330004, Jiangxi Province, China

^d College of Traditional Chinese Medicine, Jiangxi University of Chinese Medicine, Nanchang 330004, Jiangxi Province, China

^e School of Nursing, Jiangxi University of Chinese Medicine, Nanchang 330004, Jiangxi Province, China

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ABSTRACT

Background: Ulcerative colitis (UC) is an autoimmune disease that is highly susceptible to recurrence, which is still a lack of effective drugs with minor side effects in clinic. Intervention of inflammatory differentiation of dendritic cells (DCs) might be an effective strategy to treat UC. Sishen Pills (SSP) is a classic Chinese herbal formula which has been demonstrated the protective effect of UC, but the mechanism remains unclear.

Purpose: To elucidate the protective effects of SSP against UC in mice and reveal its regulatory mechanism of DCs and the key active ingredients for the UC treatment based on transcriptomics, network pharmacology and experiments validation *in vivo* and *in vitro*.

Method: The key active ingredients of SSP were detected and screened integrating LC-MS/MS and network pharmacology. A mouse UC model was induced with 3% sodium dextran sulfate and treated with SSP for 14 days to evaluate the efficacy. ELISA was used to detect the levels of IL-6, IL-1 β and TNF- α in the colon; flow cytometry was used to detect the expression levels of DCs and their subpopulations; whole transcriptomic sequencing of differential RNAs in the colon and RT-PCR to detect key miRNAs to verify the sequencing results. Mouse bone marrow-derived dendritic cells (BMDCs) were isolated, an inflammatory model was constructed using 100 ng/ml LPS, and the effects of SSP on DC proliferation and apoptosis and their surface co-stimulatory molecule expression were examined; IL-6, IL-1 β , TNF- α levels were measured by ELISA; RT-PCR and WB were performed to detect *miR-505–3p*, *CDH1*, E-cadherin expression. BMDCs with low expression of *miR-505–3p* were constructed by lentiviral transfection for further validation. The potential key ingredient was re-validated *in vivo* and *in vitro* experiment.

Results: Animal experiments showed that SSP alleviated DSS-induced UC symptoms and colonic pathological injury in mice, and inhibited IL-6, IL-1 β , TNF- α secretion and inflammatory DC proliferation and activation maturation. Network pharmacology predicted that evodiamine, isobavachalcone, curcumin, and engenol may play a key role in SSP. RNA sequencing revealed that *miR-505–3p*, as the differential miRNA, shared a large number of transcription factors with E-cadherin, and was involved in inflammatory differentiation regulation. *In vivo* experiments confirmed that SSP accelerated apoptosis, slowed down proliferation, inhibited inflammatory differentiation and IL-6, IL-1 β , and TNF- α secretion in BMDCs, and decreased *miR-505–3p*, *CDH1*, and E-cadherin levels. After knocking down *miR-505–3p*, SSP could not regulate the inflammatory differentiation and IL-6, IL-1 β ,

Abbreviations: BCA, bicinechoninic acid; BMDC, bone marrow-derived dendritic cells; CCK8, cell counting kit-8; CDH1, e-cadherin 1; DC, dendritic cells; DEGS, differentially expressed genes; ELISA, enzyme linked immunosorbent assay; EVO, evodiamine; GM-CSF, granulocyte/macrophage colony-stimulating factor; GO, gene ontology; IBD, inflammatory bowel disease; IC50, half-maximal inhibitory concentration; IL, interleukin; KEGG, Kyoto Encyclopedia of Genes and Genomes; LPS, lipopolysaccharide; miRNA, Micro RNA; OB, fecal occult blood; SMILES, Simplified Molecular Input Line Entry System; SSP, Sishen Pills; TCM, traditional Chinese medicine.

* Corresponding author.

** Corresponding authors at: Formula-Pattern Research Center of Jiangxi, University of Chinese Medicine, Nanchang, Jiangxi Province 330004, China.

E-mail addresses: 20040791@jxutcm.edu.cn (S. Cheng), 20050852@jxutcm.edu.cn (H. Zhao), 20050850@jxutcm.edu.cn (D. Liu).

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TNF- α level in BMDCs. Additionally, evodiamine was found and verified to be the key active ingredient of SSP in preventing the inflammatory differentiation of DCs.

Conclusion: SSP prevented the inflammatory differentiation of DCs by downregulating the expression of *miR-505-3p*, in which Evodiamine may played a key role.

Introduction

Ulcerative Colitis (UC) is a global chronic inflammatory bowel disease characterized by long-term and recurring irritation, inflammation, and ulceration of the lining of the colon, and is defined by the World Health Organization as one of the modern refractory diseases (Kaplan, 2015). Worldwide, the prevalence of UC is about 5 million cases, and the trend is rising year by year, UC patients are under huge physical pain and social and psychological pressure, but also greatly increase the global economic burden (Jairath and Feagan, 2020; Le Berre et al., 2023). A wide variety of treatment modalities for UC are currently available in the clinic, including pharmacological treatment with immunomodulators and biologics, as well as surgical treatment with resection of the diseased colon, and although this has greatly improved the treatment options for UC, some limitations still exist. It has been reported that a large percentage of patients do not respond effectively to biologics, and some of the drugs have more serious side effects (Danese et al., 2016; Heleig et al., 2020; Lichtenstein et al., 2015). Therefore, it is urgent to develop an alternative safe and effective new strategy.

The etiology of UC is complex, including genetics, environment and immune system abnormalities. Dendritic cells (DCs), as the most potent antigen presenting phagocytes, play a crucial role in inducing immune response and immune tolerance. Considerable evidence exists in humans and mouse models of inflammatory bowel disease (IBD) that dysfunctional DCs are capable of triggering a series of immune over-responses that contribute to the formation and exacerbation of IBD through the continuous presentation of antigens and the release of pro-inflammatory cytokines and chemokines (Rescigno and Di Sabatino, 2009). For example, in the UC mouse model, upon stimulation by pathogenic microorganisms, mature DCs greatly expanded, causing their surface co-stimulatory molecules (e.g., CD40, CD80, and CD86) to show higher expression levels, which could indirectly promote the growth and differentiation of Th17 cells releasing a large number of inflammatory cytokines and exacerbating intestinal inflammation (Lichtenstein et al., 2015). Therefore, blocking the differentiation of DCs toward an inflammatory state may be one of the effective strategies to treat UC.

Sishen Pills is a classic herbal formula for diarrhea treatment in traditional Chinese medicine (TCM), first recorded in the medical classics of "Zheng Zhi Zhun Sheng" in the Ming Dynasty (1602 A. D), consisting of six herbs: Psoraleae Fructus (*Psoralea corylifolia* L.), Myristicae Semen (*Myristica fragrans* Houtt.), Schisandrae Chinensis Fructus (*Schisandra chinensis* (Turcz.) Euodiae Fructus (*Euodia rutaecarpa* (Juss.) Jujubae Fructus (*Ziziphus jujuba* Mill.), and Zingiberis Rhizoma Recens (*Zingiber officinale* Rosc.). According to TCM theory, the symptoms of UC belong to the category of "diarrhea". Therefore, SSP is often used as a common alternative treatment for UC in China, and included in the international clinical practice guideline on the use of traditional Chinese medicine for UC (Zhang et al., 2024b). Many clinical evidence suggests that SSP improves patients' immune function and reduces the intestinal inflammatory response, as well as repairing mucosal damage and promoting its healing (Shen et al., 2021, 2018; Su et al., 2023).

In our previous study, SSP was shown to effectively decrease the number of inflammatory DCs and alleviate UC (Chen et al., 2020). However, the specific mechanism of and active compounds of SSP to effectively regulate DCs still remain unclear. Therefore, in the present study, the liquid mass spectrometry (LC-MS/MS) was used to characterize the components of SSP and predict its key components and possible targets of action through network pharmacology, meanwhile, whole transcriptomics was used to comprehensively analyze the

regulation of SSP on the immune system in UC mice, and searched for key targets of SSP inhibited inflammatory DCs. In addition, the interaction between the target compounds and the key targets was evaluated by molecular docking. The action target and potential effective constituent of SSP inhibited inflammatory to relieve the pathological colonic injury from experimental colitis induced by DSS were verified in *vivo* and *in vitro* experiments.

Materials and methods

Identification of major chemical components of SSP by quantitative liquid chromatography tandem-mass spectrometry analysis (LC-MS/MS)

Liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis was performed to identify the components of SSP using a Triple TOF-6600 mass spectrometer from AB Sciex and an LC20 ultra-high performance liquid chromatograph from Shimadzu. The mobile phases (A: 0.1% formic acid in water; B: 0.1% formic acid in acetonitrile) were delivered at 0.4 ml/min, and the instrumental column temperature was 40 °C. The liquid chromatographic separation was carried out for 14 min for each sample, with the linear gradient steps as follows: 0–11 min (95%A-5%B), 11–12 min (10%A-90%B), and 12–14 min (95%A-5%B). After the mass spectra were downlinked, the raw data were converted to mzML format by ProteoWizard, and the peaks were extracted, aligned, and corrected for retention time using the XCMS program. The peak area was corrected using the "SVR" method, and the peaks with >50% missing in each group of samples were filtered. After calibration, metabolite identification information was obtained by searching laboratory self-built database, integrating public database, AI prediction database and metDNA method.

Identification of major chemical components of SSP medicated serum by quantitative liquid chromatography tandem-mass spectrometry analysis (LC-MS/MS)

6 healthy Balb/c mice (7–8 weeks, 22±2 g weight) were randomly divided into 2 groups: blank serum and SSP medicated serum, with 3 mice in each group. Mice in the SSP medicated serum group were given 2.5 g/kg SSP once a day for 7 days. Mice in the blank serum group were given the same volume of water at the same time. 2 h after the last administration, mice blood was collected and serum was obtained after centrifugation.

The data acquisition instrument system mainly includes ultra-high performance liquid chromatography (Vanquish, UPLC, Thermo, USA) and high resolution mass spectrometry (Q Exactive HFX, Thermo, USA). The liquid chromatography parameters: Column: Waters HSS T3 (100×2.1 mm, 1.8 μ m); Mobile phase: 0.1% formic acid-water solution in phase A, 0.1% formic acid-acetonitrile-isopropanol in phase B; Flow rate: 0.3 ml/min; Column temperature: 40 °C; Injections: 2 μ L; Elution gradient: 0.0–2.0 min A/B (90:10 V/V), 6.0–15.0 min A/B (40:60 V/V), 15.1–17.0 min Water/acetonitrile (90:10 V/V), 15.1–17.0 min Water/acetic acid (90:60 V/V), 15.1–17.0 min Water/acetonitrile (90:60 V/V), 15.1–17.0 min 15.0 min A/B (40:60 V/V), 15.1–17.0 min water/acetonitrile (90:10 V/V). The Mass Spectrometry Conditions: A Q Exactive HFX high-resolution mass spectrometry system from Thermo, USA, was used for the acquisition of primary and secondary spectra. The conditions of Electrospray ionization (ESI) were as follows: sheath gas 40 psi; auxiliary gas 10 psi; ion spray voltage 3000 V/–2800 V; temperature 350 °C; ion transfer tube temperature 320 °C. The scanning

mode was Full-scan MS2 mode; the scanning mode was positive/negative ion. The first level scanning range (scan *m/z* range): 70–1050 Da, first level resolution 70,000, second level 17,500.

DSS induced UC model and drug intervention

Animals

Male SPF-grade Balb/c mice (7–8 weeks, 22±2 g weight), were purchased from Slake Jinda (Changsha, Hunan, China, certificate of quality No. SCXK (Xiang) 2019-0004) for animal experiments of SSP treatment of UC. 60 male SPF-grade C57BL/6 mice, (7–8 weeks, 21±2 g weight), were purchased from GemPharmatech (Jiangsu, China, certificate of quality No. SCXK (SU) 2018-0008) for animal experiments of EVO treatment of UC. According to the requirements of the Animal Ethics Committee of Jiangxi University of Chinese Medicine, mice were kept in the SPF grade animal room of the Animal Science and Technology Experiment Center (License No. SYXK(Gan)2022-0002), and the experiments were approved by animal ethics committee (Approval No. JZLLSC20220420 and JZLLSC20230730, approval date: Mar. 15th, 2022 and Oct. 26th, 2023). All animal experiments were conducted in a feeding environment with constant temperature and humidity, with free access to water and food for 24 h. The experiments were started after 1 week of adaptive feeding.

DSS induced UC model and drug treatment

The UC model was induced by DSS according to the literature(Wirtz et al., 2017) and all conducted based on the following procedures: the Normal group drank distilled water freely, and the other groups were free to use 3% DSS (w/v) solution for 7 days, then replaced with distilled water for another 7 days, and finally given 2% DSS (w/v) solution again for 7 days to induce the UC recurrence. From day 1, the stool of all mice was tested using the fecal occult blood (OB) test (b210102, Besso, China) according to the manufacturer’s instructions. When the OB test score was 2 (fecal occult blood positive) (Table 1), or significant anal bleeding was observed, which indicated that the model was successfully induced.

Firstly, we explored the effective dose of SSP and conducted a control experiment with the commonly used clinical drug of UC, mesalazine (5-ASA), as a positive drug. 60 mice were randomly divided into 6 mice with 10 in a group: Blank control group (Normal), model group (3% DSS), drug treatment group (DSS+SSP-H, 5 g/kg SSP, DSS+SSP-M, 2.5 g/kg SSP, DSS+SSP-L, 1.25 g/kg SSP, DSS+5-ASA, 300 mg/kg 5-ASA).

Subsequently, we selected 2.5 g/kg as the optimal dose of SSP for the mechanism exploration experiment. The mice were randomly divided into 10 mice per group: Blank control group (Normal), model group (DSS), blank+administration group (Nor+SSP, 2.5 g/kg), drug treatment group (DSS+SSP, 2.5 g/kg or DSS+EVO—H, 80 mg/kg EVO, DSS+EVO-M, 40 mg/kg EVO, DSS+EVO-L, 20 mg/kg EVO, DSS+5-ASA, 300 mg/kg 5-ASA).

Sishen Pills (SSP), purchased from Tongrentang Natural Medicine (Tangshan) Co. Ltd. (Z13020656, Beijing, China). SSP were ground into fine powder, prepared into a suspension with pure water, shaken overnight at 37 °C to fully dissolve and then used on the next day. Evodi-amine (CAS: 518–17–2), purchased from Biopurify Technology Development Co., Ltd (BP0579, Chengdu, China), configured with 0.5%

sodium carboxymethyl cellulose according to the set dose. On day 8, the mice were administered the corresponding drug by gavage at the set dose every day for a total of 14 days. Body weight, fecal consistency and OB scores were observed and calculated based on the scoring system (Table 1).

Sample collection and h&e staining

On day 21, all mice were weighed and deep anesthetized with 3% pentobarbital sodium by intraperitoneal injection according to body weights, and then sacrificed by cervical dislocation. The spleen, colon, and mesenteric lymph nodes were isolated on ice. The spleens were weighed and recorded to calculate the spleen weight index [(Spleen weight index=spleen weight/body weight) × 100%]. The mesenteric lymph nodes were collected in EP tubes filled with ice-cold RPMI-1640 and stored briefly at 4 °C for subsequent flow cytometry. The colon length was measured and photographed, followed by washing the contents with ice-cold PBS, and their weights and were recorded after absorbing excess water with filter paper to calculate the colon weight index [(colon weight index=colon weight/body weight) × 100%].

Tissue samples of colon were collected from the junction of ileum and cecum nearly 2 cm. After fixation with 4% paraformaldehyde for 72 h, the samples were embedded in paraffin and made into 4-μm-thick tissue sections. Then, conventional HE staining was performed and the sections were sealed with neutral gum. The histopathological changes of the colon were observed under an optical microscope. The pathological injury scores were evaluated according to Table 2 (Wirtz et al., 2017)

Culture, drug treatment and lentiviral vector infection of bone marrow-derived dendritic cells (BMDCs)

Isolation and culture of BMDCs

Referring to the method of Inaba et al. and modified slightly. 2 C57BL/6 male mice (SPF-grade, aged 6–7 weeks) were sacrificed and their femurs, humeri, and fibulae were isolated in a sterile environment, cleaned by ice-cold PBS, and then the bone marrow was collected by rinsing with RPMI-1640 medium, and the cell suspensions were filtered through a 70 μm cell filtration sieve to remove muscle and tissue debris. The cells were incubated using erythrocyte lysate (555,899, BD, USA) for 2 min, followed by washing the cells using RPMI-1640 twice (centrifugation at 1000 rpm for 3 min). Finally, the cells were resuspended using mouse bone marrow-derived dendritic cells (immature) complete medium (containing 20 ng/ml IL-4 and GM-CSF, CM-M151A, Procell, Wuhan, China). 72 h later, all medium was replaced and suspended cells were removed, retaining the adherent cells. By adding fresh complete medium, cells were cultured under the same conditions till day 5. On the 5th day, replaced half amount of medium with fresh medium, retaining suspended cells as much as possible and continuing the culture to the 7th day. At last, all suspended cells were collected on day 7, which was regarded as the enriched BMDCs. The purity of BMDC was identified to be more than 80% for subsequent experiments (Supplementary Fig. 1–2).

Table 1
Scoring system of a disease activity index (DAI).

Score	Weight loss	Stool consistency	Blood
0	None	Normal	Negative hemocult
1	1–5%	Soft but still formed	Negative hemocult
2	6–10%	Soft	Positive hemocult
3	11–18%	Very soft; wet	Blood traces in stool visible
4	>18%	Watery diarrhea	Gross rectal bleeding

Table 2
Scoring system for inflammation-associated histological changes.

Score	Tissue damage	Lamina propria inflammatory cell infiltration
0	None	Infrequent
1	Isolated focal epithelial damage	Increased, some neutrophils
2	Mucosal erosions and ulcerations	Submucosal presence of inflammatory cell clusters
3	Extensive damage deep into the bowel wall	Transmural cell infiltrations

Drug treatment of BMDCs

SSP was grinded to powder in a sterile environment and dissolved in sterile PBS at a concentration of 1 g/ml. Then the solution was water bathed at 65 °C for 30 min for further dissolution, and after the solution cooled naturally, it was filtered through a 200 mesh filter and a 0.22 µm filter for subsequent cellular experiments. Based on the results of the CCK8 cytotoxicity assay, the IC₅₀ of SSP on BMDCs was calculated (Supplementary Fig. 3), and the concentrations of SSP at 6, 4, and 2 mg/ml were selected as the dosages for the 3 experimental groups, namely, LPS+SSP-H, LPS+SSP-M, and LPS+SSP-L. After 20 h of SSP pre-administration, 100 ng/ml LPS was added to BMDCs to stimulate for 4 h. The treated cells were then collected for subsequent experiments.

Construction of BMDCs low-expressed miR-505-3p by lentiviral vector infection

The construction and production of *mmu-miR-505-3p* knockdown lentiviral vector was done by Genechem Co., Ltd (Shanghai, China). The viral titer was 7×10^8 TU/ml. BMDCs were collected and cultured in 24-well plates with 10^5 cells per well, and the lentivirus was added to the cell culture medium and infected for 72 h before being used for subsequent experiments. As verified by PCR and WB, lentiviral infection successfully reduced the expression of *miR-505-3p* on BMDCs (Supplementary Fig. 4).

Infected BMDCs were divided to 3 groups: Vehicle, LPS and LPS+SSP. The Vehicle group was infected by CON313 as a negative control, the LPS group and LPS+SSP group were added by 100 ng/ml LPS after infection of 68 h, besides, the LPS+SSP group was treated by 4 mg/ml SSP after infection of 48 h. Then, the treated cells were collected for the subsequent experiments.

Detection of inflammatory cytokines by ELISA

200 mg of colon tissue was accurately weighed, and RIPA lysis buffer was added according to the weight volume to completely lyse the tissue. The supernatant was collected after centrifugation for the determination of inflammatory cytokine content in colon tissue. For *in vitro* experiments, after the drug treatment, BMDCs were collected and counted, so that at least 3 samples were prepared for each experimental group, and the number of cells in each sample was not less than 5×10^5 . The cell culture medium was removed by centrifugation, and then washed by PBS, the supernatant was discarded by centrifugation. Then 100 µl of RIPA lysis buffer was added to fully lyse the cells. After centrifugation, the supernatant was transferred to a new tube for detection. According to the operating procedures of the ELISA kit, the content of TNF-α (88-7324-88, Thermo Fisher), IL-1β (88-7013-88, Thermo Fisher) and IL-6 (88-7064-88, Thermo Fisher) in the colon or BMDCs was detected. Afterwards, the absorbance value was measured at a wavelength of 450 nm using a microplate reader.

Network pharmacology analysis

Prediction of SSP targets and UC related targets

According to the composition of SSP, the chemical ingredients of *Psoraleae Fructus*, *Myristicae Semen*, *Schisandrae Chinensis Fructus*, *Euodiae Fructus*, *Jujubae Fructus*, and *Zingiberis Rhizoma Recens* were derived from the databases of Symmap (<http://www.symmap.org/>), ETCM2.0 (<http://www.tcmip.cn/ETCM2/front/#/>) and TCMSP (<http://old.tcmsp-e.com/tcmsp.php>), and matched with the compounds identified by LC-MS/MS to get the representative components of SSP. The SMILES numbers were obtained by searching the pubchem database (<https://pubchem.ncbi.nlm.nih.gov>). Then, importing the SMILES numbers into the Swiss database to predict the effective targets (<http://www.swisstargetprediction.ch>).

All UC-related gene sets were downloaded from Genecard (<https://www.genecards.org/>), OMIM (<https://www.omim.org/>), and DisGeNET (<https://www.disgenet.org/>) to obtain the corresponding disease gene symbol information. Based on all the screening results from databases, repetitive genes were proposed as the disease targets related to UC. The targets of representative components of SSP and the UC-related targets were cross-analyzed by Venny (<https://bioinfogp.cnb.csic.es/tools/venny/>) and the crossover results were used as potential key targets for SSP in the treatment of UC.

Network analysis and functional enrichment analysis

Import the screened potential key targets into the STRING database, select "Homo sapiens" as the species, and set the scoring criteria to >0.90. Remove discrete targets, then obtain a protein-protein interaction (PPI) network and save it as a TSV format file. The TSV file was imported into Cytoscape 3.10.0 software, and the topology of the network was calculated using the plug-in Centiscape 2.2, and the core regulatory targets were screened to construct the PPI interaction network. Meanwhile, the core regulatory targets were imported into Sangerbox platform (Shen et al., 2022) for GO and KEGG functional enrichment analysis. Finally, a "drug-components-targets-signaling pathways" network relationship was constructed of the obtained key components of SSP, potential key targets for SSP in the treatment of UC, and the enriched signaling pathways, and then visualized using Cytoscape 3.10.0.

Molecule docking

Based on the topological parameters, the top-ranked macromolecules and small molecules were selected for molecular docking to predict their therapeutic potential. The drug small molecules were also selected with reference to the quality identification standard of Chinese medicines in the Chinese Pharmacopoeia (2020 edition). The small molecule structures were obtained from PubChem database, which 3D structural formulas were drawn by Chemdraw software. The protein crystal structure of the core target was obtained from the PDB database (<https://www.rcsb.org/>), and the selection principles were as follows: human origin protein, high resolution (less than or equal to 2.5 Å), with original ligands, and the crystallization pH value should be as close as possible to the normal physiological range of the human body. Finally, protein receptors and small molecule ligands were processed for docking using AutoDock vina, and the results were visualized with PyMOL.

Transcriptomic analysis

RNA extraction and sequencing

Extract total RNA from tissues according to the manufacturer's instructions and remove genomic DNA. Verify RNA quality using 2100 biological analyzer. Construct a sequencing library using only high-quality RNA samples. A RNA seq transcriptome library was prepared using 5 µg of total RNA, quantified by TBS380, and subjected to deep sequencing by Shanghai Majorbio Bio Pharm Biotechnology Co., Ltd. (Shanghai, China). Data analysis and mapping were performed using the Meijibio platform.

Differential expression analysis and functional enrichment

To identify differentially expressed mRNAs and miRNAs between the two different groups, the expression level of each transcript was calculated based on the transcripts per million reads (TPM) method. RSEM (<http://deweylab.biostat.wisc.edu/rsem/>) was used to quantify gene abundance. Differential expression analysis was performed using DESeq2 with $\text{p.adjust} \leq 0.01$, and DEGs with a fold change of >4.0 or <-4.0 were considered to express significantly different mRNAs.

similarly, DEGs with expression with $\text{padjust} \leq 0.01$, and a fold change of >2.0 or <-2.0 were considered to express significantly different miRNAs. In addition, functional enrichment analysis, including GO and KEGG.

Detection of mRNA and miRNA by qPCR

50–100 mg of colon tissue (or not less than 5×10^5 BMDCs) was taken for mRNA and miRNA detection, respectively. Total RNA was isolated using the Universal RNA Mini Kit (MF036–01, Mei5bio, China) according to the instructions, reverse transcription was performed using the Sprint-qPCR RT kit (MF949–01, Mei5bio, China), and finally the fluorescence signal intensity of samples was detected using a $2 \times$ real-time PCR super mix.

Similarly, miRNA was extracted using the Animal miRNA Extraction Kit (MF044–01, Mei5bio, China) according to the instructions, followed by reverse transcription using microRNA first strand synthesis kit (tail-addition method) (MF283–01, Mei5bio, China), finally, miRNA fluorescence quantitative PCR detection kit (MF307–01, Mei5bio, China) was used for quantitative detection. The PCR cycle conditions were set as follows: 94 °C for 3 min; then, 40 cycles were amplified at 94 °C for 20 s, 60 °C for 20 s, and 70 °C for 20 s. Circulating threshold (CT) values were obtained from the CFX96 real-time PCR system (Bio-Rad, USA) and standardized to GAPDH levels. The relative mRNA and miRNA levels of each sample were measured by the $2^{-\Delta\Delta CT}$ method. The primer sequences are listed in Table 3 below.

Flow cytometry

The lymph node single-cell suspension was prepared as follows: The collected mesenteric lymph nodes were placed in a petri dish, lightly ground and squeezed with 1 ml of RPMI-1640 cell medium on ice by using a 5 ml sterile syringe. The cell suspensions were collected and filtered with a 70 μm cell sieve to remove impurities. After centrifugation ($300 \times g$, 10 min), the supernatant was discarded and the cells were washed twice with 1 ml of staining buffer. Cells were subsequently blocked for 10 min with 0.5 μl of FC Block (C0161021920353, Tonbo Biosciences). Calculate the amount of surface antibody for the single cell suspension and prepare a cocktail, including: APC–Cy7 anti-mouse CD11c (561,241), APC anti-mouse CD103 (562,772), AF488 anti-mouse E-cadherin (563,570), BV510 anti-mouse SIRP α (740,159), PE anti-mouse F4/80 (565,410), PE–Cy7 anti-mouse Ly6c (560,593), BV510 anti-mouse CD24 (747,717), PE anti-mouse MHC-II (557,000), BV421 anti-mouse Flt3 (562,898) from BD Biosciences, BV421 anti-mouse FcyRI (139,309), PE–Cy7 anti-mouse CD115 (135,523), PE anti-mouse Fc ϵ RI α (134,307) from BioLegend. Then the prepared cocktail was added into the sample and mixed, and incubated at 4 °C for 30 min in the dark. After incubation, centrifuge ($350 \times g$, 5 min) the cells and add 1 ml stain buffer to wash twice. After the last centrifugation, the supernatant was discarded and 0.5 ml stain buffer was added to resuspend the cells for detection.

For *in vitro* experiment, at the end of the drug treatment, BMDCs cells were removed from the medium, washed once with PBS, and then collected and counted, with no less than 5×10^5 cells per sample. Ki67 (KGA325, KeyGene Biotech, Jiangsu, China) and Annexin V (AP101–100-AVF, Multi Sciences, Hangzhou, China) were detected according to the kit instructions. Surface antibody staining was performed according to flow cytometry of mononuclear lymphocytes, the antibodies were used as follow: APC–Cy7 anti-mouse CD11b (88–7324–88) from Thermo Fisher, PerCP–Cy5.5 anti-mouse CD11c (560,584), PE–Cy7 anti-mouse CD86 (561,241), AF488 anti-mouse E-cadherin (563,570), APC–Cy7 anti-mouse TNF- α (560,658), PE anti-mouse MHC-II (566,776) from BD Biosciences. Prior to detection, antibody fluorescent single-stain tubes were set up for compensatory adjustment, and isotype negative control samples were set up for precisely gating. All data were analyzed by FlowJo 10.4.

Western blot

Protein supernatant was prepared as described above. A small amount of protein supernatant was taken and the protein concentration was determined by the BCA method. For western blot detection, 30 μg of protein was subjected to 10% SDS-PAGE and transferred to PVDF membrane and incubated overnight with primary antibody, including: Rabbit anti-GAPDH (ab181602) and Mouse anti-E-cadherin (ab76055) from Abcam. The next day, the membrane was washed with TBST and incubated with the second antibody for 2 h ECL solution was added to observe the protein bands by gel imaging system (ChemiDoc XRS+, Bio-Rad, USA). Image J software was used to analyze the images.

Statistical analysis

All data from this experiment were statistically analyzed and plotted using GraphPad Prism 9.5.1 software, and the cellular experiments were repeated at least three times for each group, so all results are presented as means $\pm$ SEM of three independent experiments. Comparisons between groups were performed using the sample *t*-test, and $p < 0.05$ was defined as statistically significant.

Results

SSP effectively alleviated DSS-induced ulcerative colitis in mice

DSS-induced model was used to verify the protective effect of SSP on UC. The DAI score is an important indicator to fully respond to colitis from the three perspectives of body weight loss, fecal consistency, and fecal occult blood. In Fig. 1, we found that different doses of SSP all had some ameliorating effect on DSS-induced colitis, with the most obvious effect of SSP at 2.5 g/kg, which was able to approach the therapeutic effect of mesalazine. Therefore, in the subsequent mechanism exploration experiment, 2.5 g/kg was selected as the research dose of SSP. From day 6, the DSS group showed significant weight loss, extensive blood in

Table 3
The primer sequences of miRNA and mRNA.

miRNA/mRNA	Forward	Reverse
miR-505-3p	GCGAGCACCGTCAACACT	Universal Reverse primer
miR-504-5p	GAGACCTGGTCTGCAC	
miR-380-5p	CGATGGTTGACCATAGAACATGCG	
miR-370-3p	GCCTGCTGGGGTGGAACCTGGT	
miR-490-3p	GCAACCTGGAGGACTCC	
miR-485-5p	AGAGGCTGGCCGTGAT	GCTCTTTGACCACCGTTCTC TCTAGACGGCAGGTCAAGTCCACC
miR-335-5p	GCGGCGGTCAAGAGCAATAACG	
miR-9-3p	GAGGCCCGTTTCTCTCTTTG	
miR-10b-3p	AGCGCACAGATTCTGATTCTAG	
CDH1	GACGTCCATGTGTGTGACTG	
GAPDH	CCACCCATGGCAAATTCATGGCA	

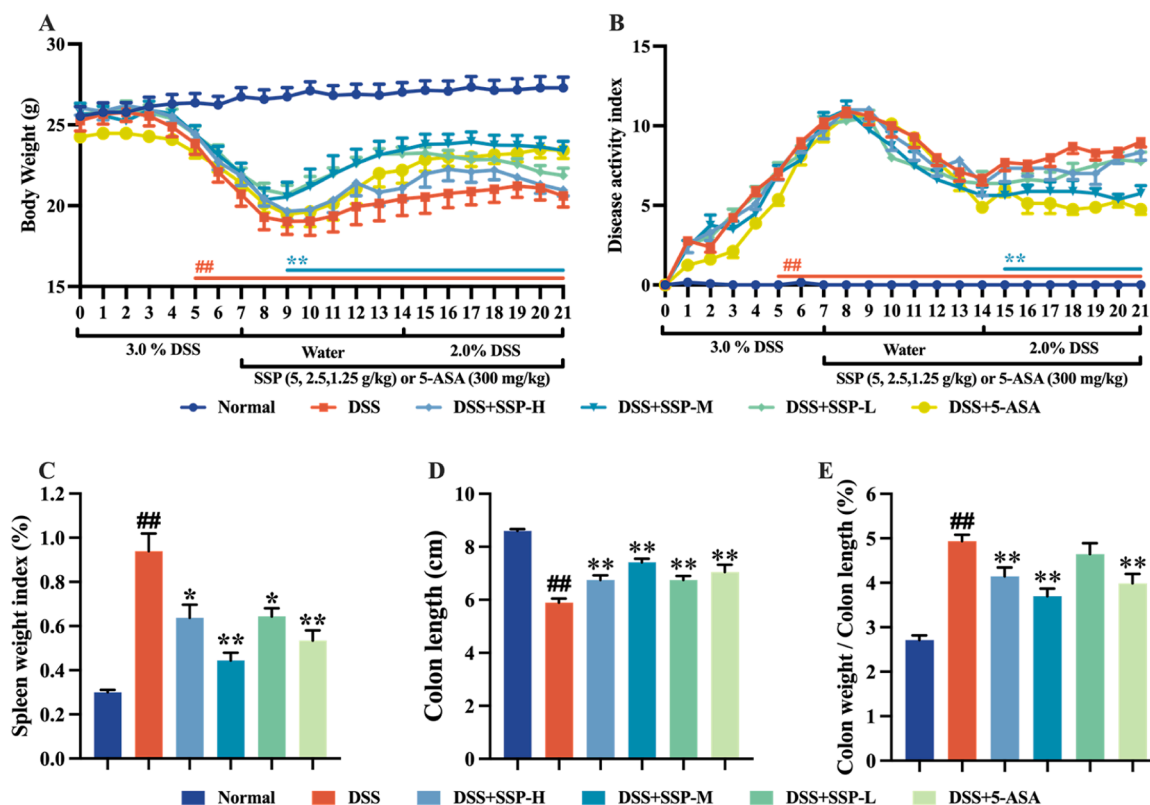


Fig. 1. Optimal dose assessment of SSP on DSS-induced UC (n = 10). (A) Body weight. (B) Disease activity score (DAI). (C) Spleen weight index. (D) Colon length. (E) Colon weight/colon length. Compared with Normal group, #*P*<0.05, ##*P*<0.01; Compared with DSS group, **P*<0.05, ***P*<0.01.

the stool, loose feces, and a significant increase in the DAI score compared with the Normal group. After continuous administration of SSP from day 7, the DSS+SSP group gradually regained their body weight, had less loose stool or blood in stool, and showed a decreasing trend in DAI score compared with the DSS group (*P*<0.05 or 0.01) (Fig. 2A-B). Additionally, the DSS group had abnormally enlarged spleens and increased splenic weight index compared with the Normal group and Nor+SSP group (Fig. 2C-D), the colon length was shortened, the colon was edematous, with the colon weight index increased (Fig. 2E-H). Meanwhile, the H&E staining results of colons were seen to be disorganized with marked inflammatory infiltration in the lamina propria, crypts disappeared, and ulcerations were apparent, and the pathohistological scores increased significantly (Fig. 2I-J), which implies that DSS successfully induced symptoms similar to those of human UC. In contrast, the DSS+SSP group recovered the spleen weight index and colon length, the colon weight and colon weight index decreased, the inflammatory infiltration and ulceration of colons was reduced by microscopy, with the pathohistological scores were significantly decreased.

To further confirm the alleviating effect of SSP on colitis, the pro-inflammatory cytokines TNF- α , IL-1 β , and IL-6 were detected in the colonic tissues using ELISA. Compared with the Normal group, the expression levels of TNF- α , IL-1 β , and IL-6 in the DSS group were significantly elevated, suggesting that colonic inflammation was occurring. Inversely, the expression levels of TNF- α , IL-1 β and IL-6 were significantly decreased in the DSS+SSP group (Fig. 2K-M). The above results indicated that SSP could effectively alleviated inflammation and the symptoms similar to those of human UC caused by DSS.

LC-MS/MS analysis of the chemical constituents in SSP

LC-MS/MS was used to identify the representative chemical compound of SSP and a total of 4973 compounds were identified from SSP,

and representative base peaks and ion flow chromatograms are shown on the left side of Fig. 3. The representative compounds identified in SSP and their associated herbs are listed in Table 4.

Component identification and network pharmacology target prediction of SSP medicated serum

Blank serum, SSP medicated serum and SSP were evaluated by LC-MS/MS in Fig. 4A-B. Among them (Fig. 4C), 97 components overlap in SSP and SSP medicated serum, but do not appear in blank serum. These components may be compound that can be absorbed into circulation and maintain a certain exposure level, and are the main pharmacodynamic substances for SSP to play a role.

The components were screened according to the Absorption rate and drug-like properties, and Percent Human Oral Absorption $\geq 30\%$ and DL (drug-likeness) ≥ 0.18 were selected as Drug target mining. They were imported into BATMANTCM platform for target gene prediction and analysis. Target gene prediction was obtained by calculating the similarity score between "blood compound - target" and "known drug - target", and the target with score ≥ 20 was set as the target gene of blood compound in the BATMAN-TCM platform.

To clarify the potential function of the above targets, gene function enrichment analysis was performed using the Sangerbox platform (Fig. 4D-E). The results of GO enrichment analysis showed that SSP treatment of UC involved the following biological functions and processes: transmembrane transporter activity, cell-cell signaling, cell death, cell proliferation, cell differentiation, response to stress. Besides, the results of KEGG enrichment analysis showed that SSP treatment of UC involved follow pathways: cell adhesion related signaling pathways (such as Gap junction, Tight junction), inflammation related signaling pathways (such as MAPK signaling pathways, AMPK signaling pathways, TGF-beta signaling pathways and inflammatory mediator regulation of TRP channels).

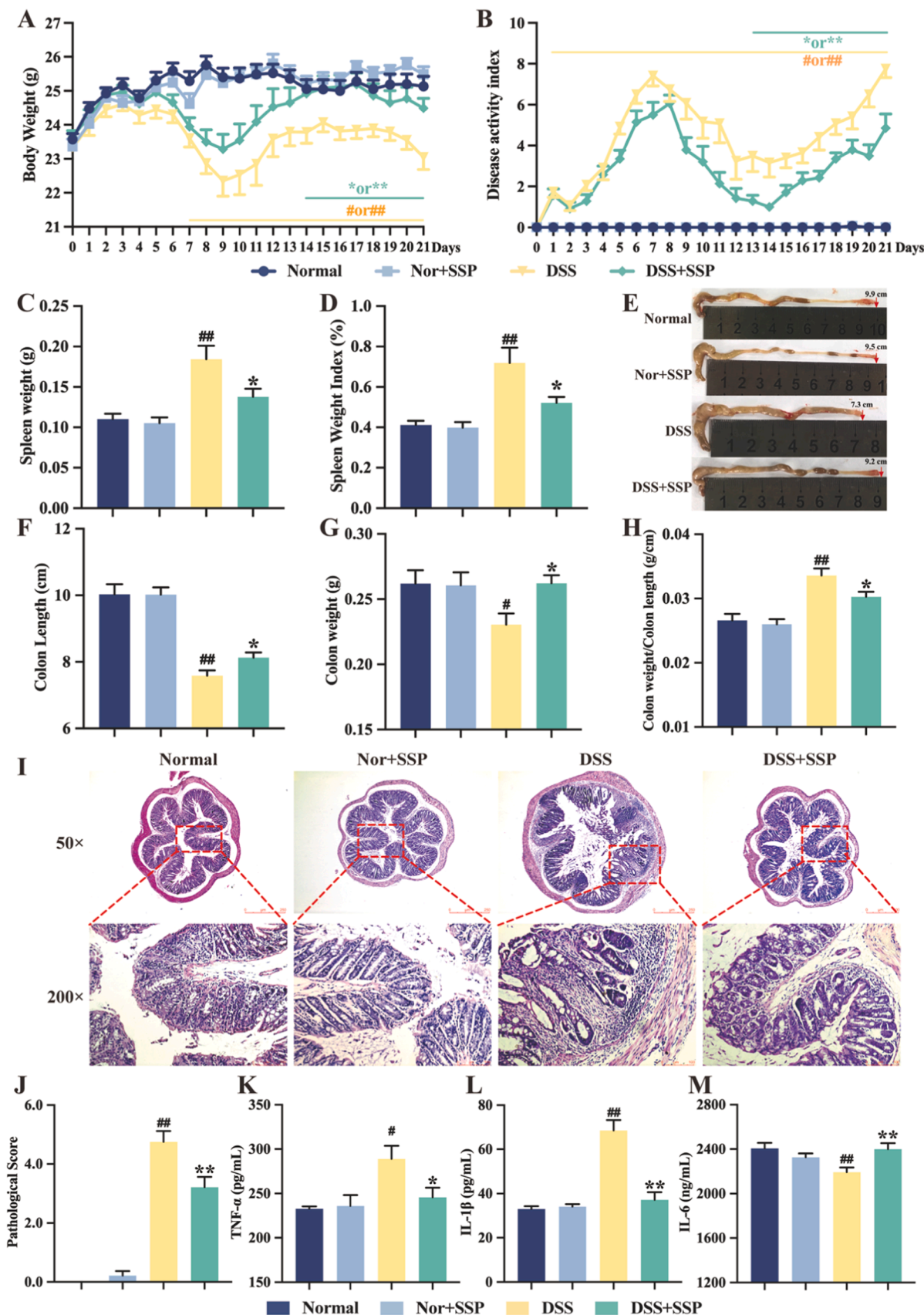


Fig. 2. Effects of SSP on DSS-induced UC ($n = 10$). (A) Body weight. (B) Disease activity score (DAI). (C) Spleen weight. (D) Spleen weight index. (E) Representative images of mice colon. (F) Colon length of mice. (G) Colon weight. (H) Colon weight/colon length. (I) Representative sections for hematoxylin and eosin (H&E) staining in each group. (J) Pathological score of colon tissue in each group. The expression levels of TNF- α (K), IL-1 β (L), and IL-6 (M) in colon tissue. Compared with Normal group, $^{\#}P < 0.05$, $^{##}P < 0.01$; Compared with DSS group, $^{*}P < 0.05$, $^{**}P < 0.01$.

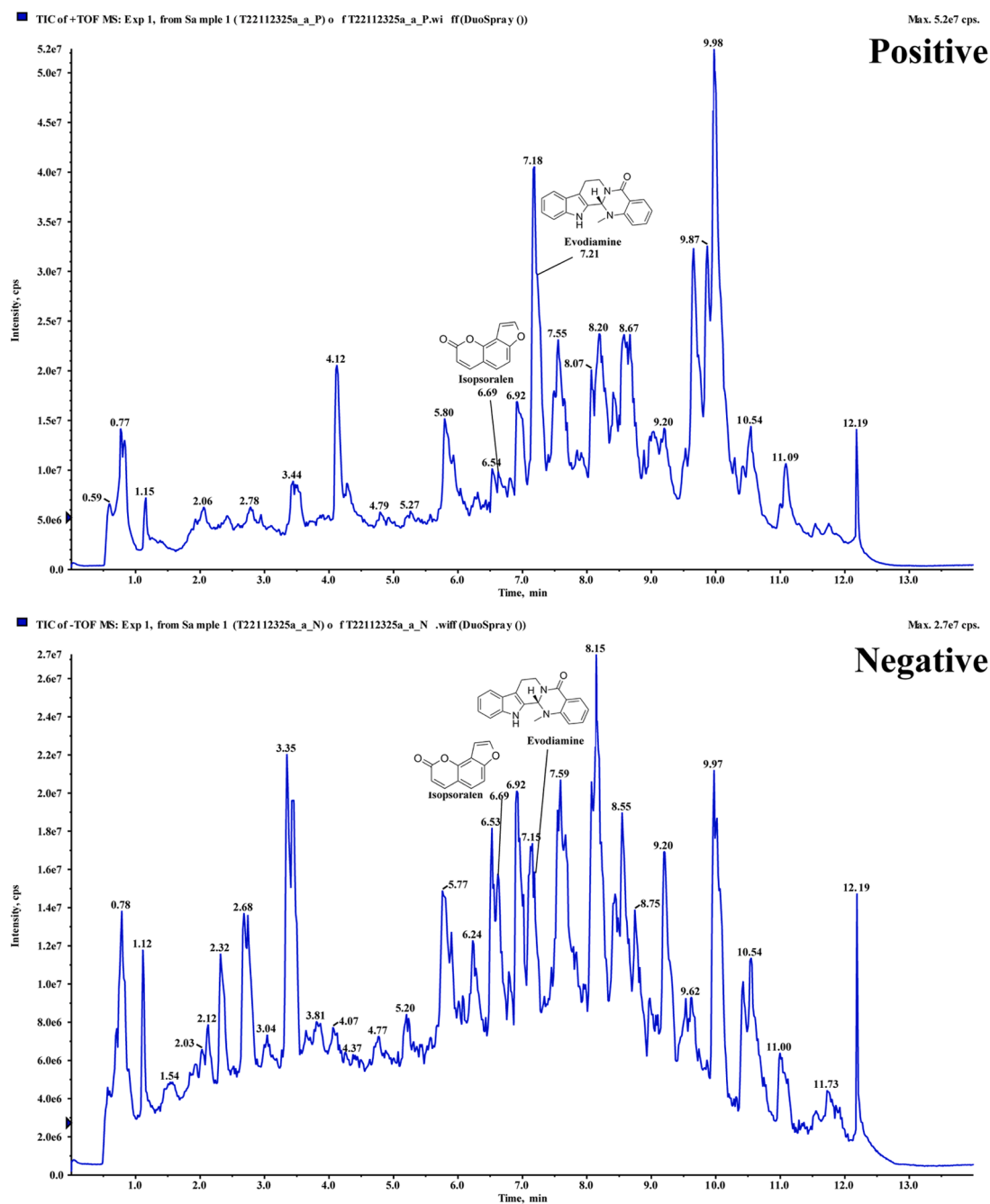


Fig. 3. Total ion chromatogram of SSP. (A) Positive ion mode. (B) Negative ion mode.

Transcriptomic analysis of SSP treating UC

Transcriptome analysis was performed by RNA sequencing (RNA-seq) in order to reveal the underlying mechanism and potential molecular targets of SSP in the effective treatment of UC. In the analysis of mRNA, compared with Normal group, there were 298 differential mRNAs in DSS group, including 201 up-regulated and 97 down-regulated; compared with DSS group, there were 18 differential mRNAs in DSS+SSP group, including 10 up-regulated and 8 down-regulated (Fig. 5A); 85 differential mRNAs were selected according to the treatment trend. In the analysis of miRNA, compared with Normal group, 42 differential miRNAs were screened in DSS group, of which 18 were up-regulated and 24 down-regulated; compared with DSS group, there were 15 differential miRNAs in DSS+SSP group, including 8 up-regulated and 7 down-regulated (Fig. 5A); 27 differential miRNAs

were screened according to the trend of treatment. The cluster heat map of 85 differential mRNA is shown in Fig. 5B. By drawing the Venn map, it was seen that the corresponding genes of the 85 differential mRNAs were highly coincident with the core targets predicted by network pharmacology, which indicated that these 85 genes were the key targets of SSP in the treatment of UC. Subsequently, these 85 key targets were analyzed for GO and KEGG enrichment. GO results showed that these genes were highly enriched in inflammatory response related signaling pathways and highly correlated with cell biological adhesion and other functions (Fig. 5C); KEGG results showed that these genes were closely related to tight junction and adhesion and were involved in multiple signal pathways that affect the activation, migration and antigen presentation function of dendritic cells (such as cAMP signaling pathways and Rap1 signaling pathways) (Fig. 5D).

Then, based on the above key mRNAs and miRNAs, the ceRNA

Table 4
Chemical components of SSP.

RT (min)	Compounds	Formula	CAS	Origin
0.6148	Geraniin	C41H28O27	60,976–49–0	Zingiberis Rhizoma Recens
0.6855	Rhein	C15H8O6	478–43–3	Myristicae Semen
0.9271	Citric acid	C6H8O7	77–92–9	Euodiae fructus
1.3397	Naringin	C27H32O14	10,236–47–2	Psoraleae Fructus
1.4358	Cycloheximide	C15H23NO4	66–81–9	Zingiberis Rhizoma Recens
1.5	Baicalin	C21H18O11	21,967–41–9 31,564–28–0	Myristicae Semen
1.5625	Gallic acid	C7H6O5	149–91–7	Myristicae Semen
1.6747	Sucrose	C12H22O11	57–50–1	Myristicae Semen
2.3072	Harpagoside	C24H30O11	19,210–12–9	Zingiberis Rhizoma Recens
2.555	Chlorogenic Acid	C16H18O9	202,650–88–2	Myristicae Semen
3.3564	Roseoside	C19H30O8	54,835–70–0	Jububae Fructus
3.5801	Bicuculline	C20H17NO6	485–49–4	Myristicae Semen
3.7786	4-Hydroxybenzaldehyde	C7H6O2	123–08–0	Psoraleae Fructus
3.7953	Rutin	C27H30O16	153–18–4	Euodiae fructus
4.1417	Ferulic acid	C10H10O4	537–98–4	Schisandrae Chinensis Fructus
4.2832	4-Hydroxybenzoic acid	C7H6O3	99–96–7	Jububae Fructus
4.9247	Cinnamic acid	C9H8O2	140–10–3	Zingiberis Rhizoma Recens
4.974	Daidzein	C15H10O4	486–66–8	Psoraleae Fructus
4.9962	Caffeine	C8H10N4O2	58–08–2	Euodiae fructus
5.7178	(-)-Catechin	C15H14O6	18,829–70–4	Jububae Fructus
5.8349	Isorhamnetin	C16H12O7	480–19–3	Euodiae fructus
6.0599	Pentadecanoic acid	C15H30O2	1002–84–2	Zingiberis Rhizoma Recens
6.1891	Limonin	C26H30O8	1180–71–8	Euodiae fructus
6.2229	Glycyrrhizic acid	C42H62O16	1405–86–3	Myristicae Semen
6.2427	Formononetin	C16H12O4	485–72–3	Myristicae Semen
6.3214	Rutaein	C26H30O9	33,237–37–5	Euodiae fructus
6.4388	Isoginkgetin	C32H22O10	548–19–6	Zingiberis Rhizoma Recens
6.6371	1-Methyl-2-nonylquinolin-4(1H)-one	C19H27NO	68,353–24–2	Euodiae fructus
6.6903	Isobavachalcone	C20H20O4	20,784–50–3 54,676–49–2	Psoraleae Fructus
6.7994	Evodol	C26H28O9	22,318–10–1	Euodiae fructus
6.9028	Zingiberis Rhizoma Recensol	C17H26O4	58,253–27–3	Zingiberis Rhizoma Recens
7.0218	Malabaricano	C20H24O5	74,683–16–2	Myristicae Semen
7.1727	Naringenin	C15H12O5	67,604–48–2	Myristicae Semen
7.2186	Evodiamine	C19H17N3O	518–17–2	Euodiae fructus
7.303	Eugenol	C10H12O2	97–53–0	Euodiae fructus
7.6386	D-Tartaric acid	C4H6O6	147–71–7	Jububae Fructus
7.7343	Melanin	C18H10N2O4	8049–97–6	Myristicae Semen
7.8267	Emodin	C15H10O5	518–82–1	Myristicae Semen
8.0542	(8)-Zingiberis Rhizoma Recensol	C19H30O4	23,513–08–8	Zingiberis Rhizoma Recens
8.1264	4-[(2S,3S,4R,5R)–5-(4-hydroxy-3-methoxyphenyl)–3,4-dimethyloxolan-2-yl]–2,6-dimethoxyphenol	C21H26O6	–	Myristicae Semen
8.8496	Quinic acid	C7H12O6	77–95–2	Schisandrae Chinensis Fructus
9.2706	3,4-Dihydroxybenzaldehyde	C7H6O3	139–85–5	Jububae Fructus
9.5162	Farnesyl acetate	C17H28O2	4128–17–0	Zingiberis Rhizoma Recens
9.765	Jubanin B	C43H47N5O6	60,375–08–8	Jububae Fructus
10.0187	Maslinic acid	C30H48O4	4373–41–5	Jububae Fructus
10.5038	Malabaricone C	C21H26O5	63,335–25–1	Myristicae Semen
10.5902	Curcumin	C21H20O6	458–37–7	Myristicae Semen
10.6801	Ursolic acid	C30H48O3	77–52–1	Jububae Fructus
10.8806	Betulonic acid	C30H48O3	472–15–1	Jububae Fructus
10.8836	Stearic acid	C18H36O2	57–11–4	Euodiae fructus
11.4236	Betulonic acid	C30H46O3	4481–62–3	Jububae Fructus
11.4331	Oleanolic acid	C30H48O3	508–02–1	Euodiae fructus

network was constructed to screen out the miRNA involved in the core regulation. To verify the reliability of the sequencing results, key regulated miRNAs were detected in colon tissues by qPCR. The expression of *miR-505-3p*, *miR-504-5p*, *miR-485-5p*, *miR-335-5p*, and *miR-9-3p* increased in the DSS group and decreased in the DSS+SSP group, whereas the expression of *miR-380-5p* and *miR-490-3p* decreased in the DSS group and increased in the DSS+SSP group rose. SSP could reverse the trend of miRNAs. The qPCR results showed the trend of relative expression of core-regulated miRNA in each group was the same as that of RNA-seq (Fig. 5e–f).

Monocyte-derived inflammatory DCs highly express E-cadherin and promote intestinal inflammation, which was reported in 2010 (Siddiqui et al., 2010). E-cadherin is a cell adhesion protein which encoded by gene CDH1. Through the transcription factor database and literature

research (Supplementary Fig. 6), *miR-505-3p* was found closely related to E-cadherin, the key marker of inflammatory DCs. Based on the above results, we hypothesized that SSP treats UC by interfering with the inflammatory differentiation of DCs by inhibiting the expression of *miR-505-3p*.

Regulation of DCs by SSP in UC mice in vivo

In order to explore how SSP regulates DCs in DSS-induced UC mice, flow cytometry was used to detect the frequency of DCs and their sub-populations in the mesenteric lymph nodes of mice. E-cadherin⁺DC is a special kind of inflammatory DCs that accumulate in the inflammatory site (Siddiqui et al., 2010). After DSS induced UC, the frequency of CD11c⁺CD103⁺E-cadherin⁺DC in DSS group increased significantly,

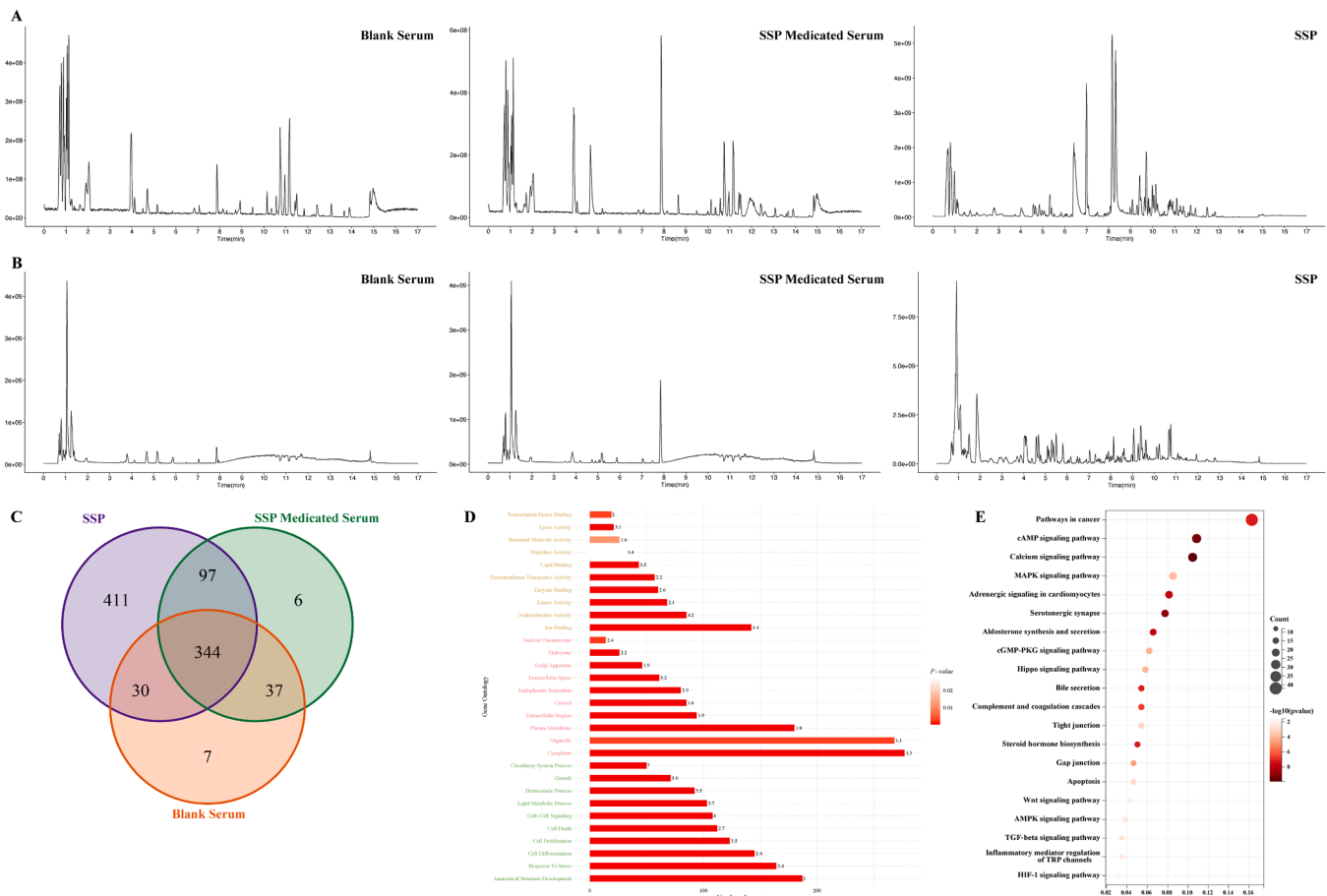


Fig. 4. Network pharmacology analysis of SSP treating UC. (A) Positive ion model of blank serum, SSP medicated serum and SSP. (B) Negative ion model of blank serum, SSP medicated serum and SSP. (C) Venn analysis of blank serum, SSP medicated serum and SSP. (D). GO enrichment analysis of 344 targets. (E) KEGG enrichment analysis of 344 targets.

while decreased significantly after SSP administration (Fig. 6A1-A2). In addition to E-cadherin, other surface markers that are highly expressed on inflammatory DCs include SIRP α , F4/80, Ly6c, MHC-II, Fc γ RI and Fc γ RI- α . Consistent with the expression trend of E-cadherin in DCs, the expression levels of SIRP α , F4/80, Ly6c, MHC-II, Fc γ RI and Fc γ RI- α in DCs were significantly increased by DSS induced inflammatory reaction (Fig. 6B1-G2). Compared with DSS group, the expression levels of these subsets in DSS+SSP group showed a significant decrease trend. SSP obviously inhibited the inflammatory co-stimulatory molecules of DCs in mice to alleviated the colonic inflammatory injury.

In addition, cell surface markers related to DCs maturation and inflammatory differentiation were also detected. Compared with the Normal group, the expression levels of CD115, CD24 and Flt3 on DCs in the DSS group increased significantly, indicating that the inflammatory environment induced by DSS caused a large number of DCs to mature and differentiate into an inflammatory state; compared with the DSS group, in the DSS+SSP group, SSP well inhibited the expression levels of CD115, CD24 and Flt3 (Fig. 7A1-C2). The above experimental results show that SSP can effectively inhibit the maturation and inflammatory differentiation of DCs to relieve UC in mice.

Regulation of DCs maturation and inflammatory differentiation by SSP *in vitro*

Effect of SSP on proliferation and apoptosis of BMDCs

Under physiological conditions, DC is small in number but widely distributed in the body, and most of them are in an inactive state. When

infection or inflammation occurs, immature DCs rapidly mature and proliferate, effectively activating initial T cells to produce antigen-specific responses, so a large number of mature dendritic cells also confirm the generation of inflammation in the body (Patente et al., 2019; Zanna et al., 2021). As shown in Fig. 8A, the Annexin V positivity rate of BMDCs in the LPS group had a more obvious increase compared with that of the Vehicle group, while LPS+SSP-H was able to further promote apoptosis of BMDCs compared with that of the LPS group ($p < 0.05$ or $p < 0.01$). In terms of inhibiting proliferation of BMDCs, the expression of Ki67 of BMDCs in the LPS group increased significantly compared with the Vehicle group; while after the intervention of different doses of SSP, the Ki67 positivity rate in each group all showed a decreasing trend, especially LPS+SSP-H and LPS+SSP-M group showed the most significant changes ($p < 0.01$) (Fig. 8B). The above results suggested that SSP could inhibit the proliferation of BMDCs and accelerate their apoptosis *in vitro*.

Effects of SSP on the expression of co-stimulatory molecules on BMDCs

When infection or inflammation occurs, mature DCs are abundant, and their most important recognition feature is the expression of costimulatory molecules (Xiao and Xia, 2023). Stimulated by LPS, BMDCs in LPS group rapidly proliferated and matured, leading to corresponding inflammation. Compared with Vehicle group, the expression levels of co-stimulatory molecules CD86 and MHC-II were significantly increased (Fig. 8C1-D2). E-cadherin⁺ DCs, as the signature inflammatory dendritic cell subpopulation, also significantly increased the expression level of E-cadherin after LPS stimulation (Fig. 8E). At the same time, the

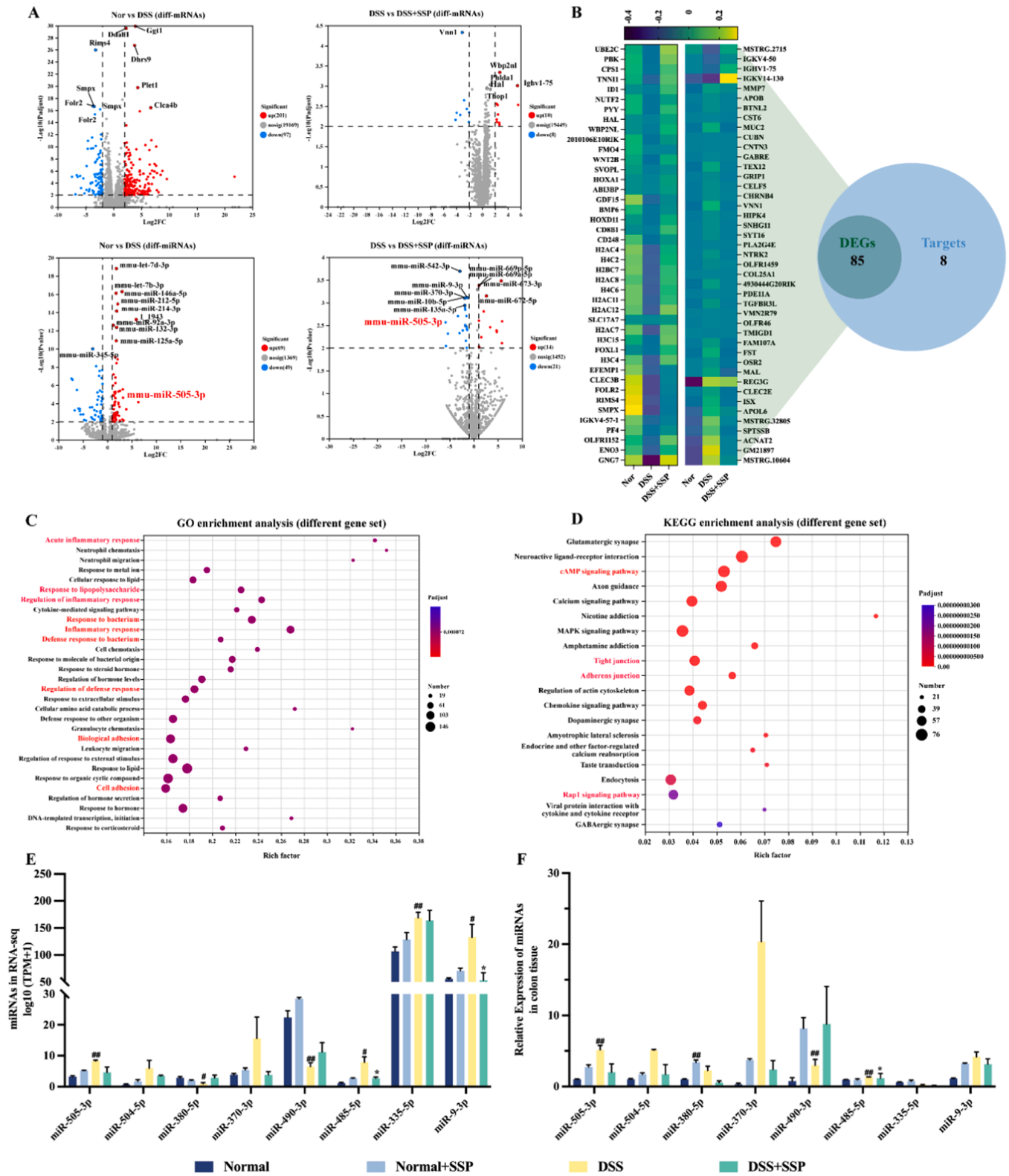


Fig. 5. Transcriptomic analysis of SSP treating UC ($n = 3$). (A) Volcano plot of differentiate mRNAs (up); Volcano plot of differentiate miRNAs (down). (B) Cluster analysis heatmap of 85 differential mRNAs (left); Venn diagram of 85 differential genes and 94 core targets analyzed by network pharmacology. (C) GO enrichment analysis of 85 differentiate genes. (D). KEGG enrichment analysis of 85 differentiate genes. (E) Relative expression of key miRNAs in RNA-seq. (F) Relative expression of key miRNAs in mice colon tissue detected by PCR. Compared with Normal group, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with DSS group, $^*P < 0.05$, $^{**}P < 0.01$.

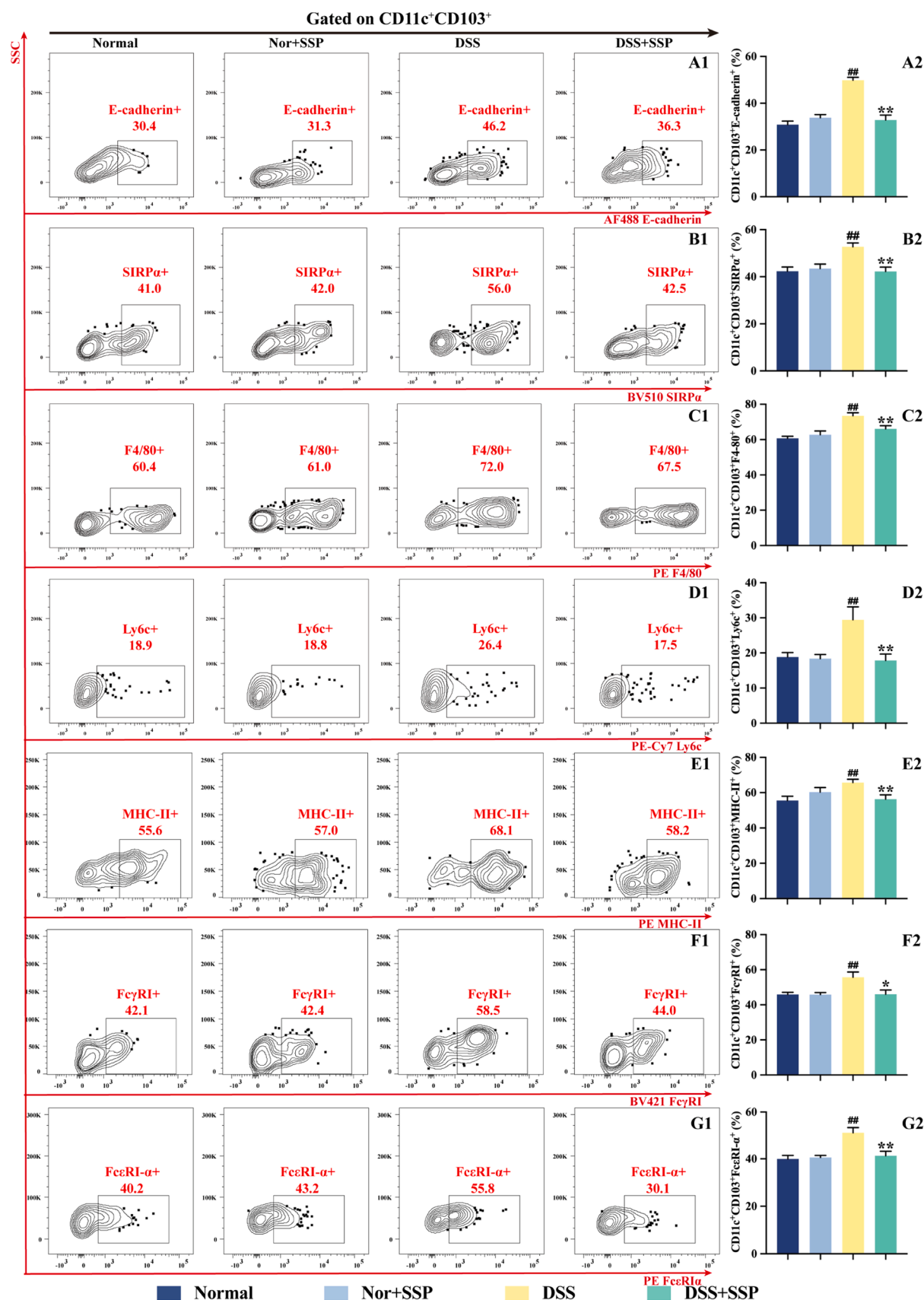


Fig. 6. Regulation of SSP on DCs in DSS induced UC mice ($n = 10$). Representative flow cytometry scatterplot of E-cadherin (A1), SIRPα (B1), F4/80 (C1), Ly6c (D1), MHC-II (E1), FcγRI (F1) and FcεRI-α (G1) in each group. Flow cytometry histograms of E-cadherin (A2), SIRPα (B2), F4/80 (C2), Ly6c (D2), MHC-II (E2), FcγRI (F2) and FcεRI-α (G2) in each group. Compared with Normal group, # $P < 0.05$, ## $P < 0.01$; Compared with DSS group, * $P < 0.05$, ** $P < 0.01$.

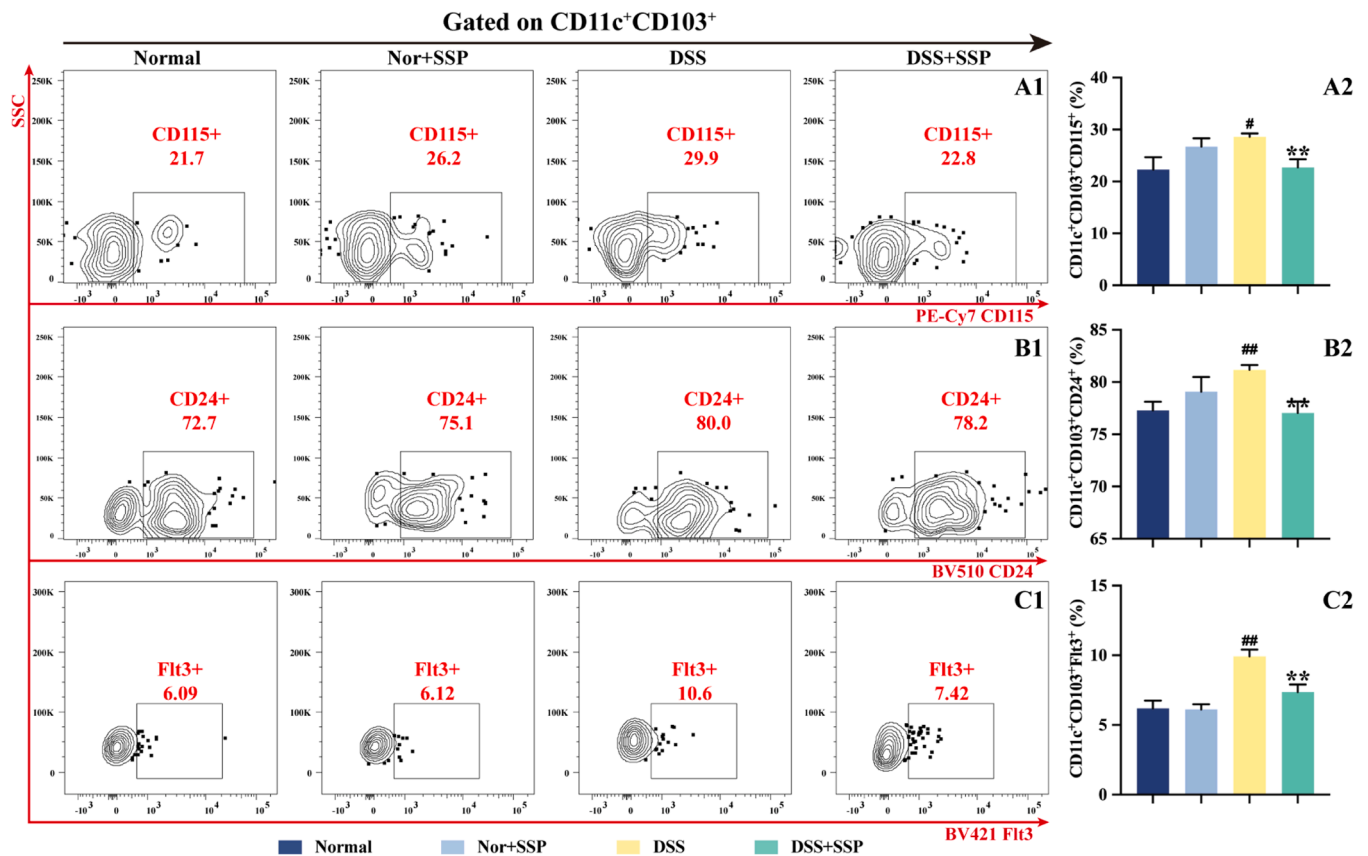


Fig. 7. Regulation of SSP on DCs in DSS induced UC mice ($n = 10$). Representative flow cytometry scatterplot of CD115 (A1), CD24 (B1) and Flt3 (C1) in each group. Flow cytometry histograms of E-cadherin CD115 (A2), CD24 (B2) and Flt3 (C2) in each group. Compared with Normal group, $^{\#}P < 0.05$, $^{\#\#}P < 0.01$; Compared with DSS group, $^{*}P < 0.05$, $^{**}P < 0.01$.

expression level of TNF- α , as a signature pro-inflammatory cytokine, increased considerably on BMDCs as well (Fig. 8F), suggesting that BMDCs was in an inflammatory state. After treatment with different doses of SSP, the expression levels of CD86, MHC-II, E-cadherin and TNF- α on BMDCs in these three SSP-treat groups were obviously decreased ($p < 0.01$), indicating that SSP could effectively inhibit the activation and maturation of BMDCs.

Effects of SSP on the secretion of inflammatory cytokines in BMDCs

DCs present antigen to naive T cells under the action of surface costimulatory molecules, release inflammatory cytokines, and participate in the regulation of immune response balance. After LPS stimulation, the levels of inflammatory cytokines IL-6, IL-1 β and TNF- α secreted by BMDCs increased markedly compared with the Vehicle group. However, different doses of SSP could reverse this trend. In the SSP-treated group, the secretion levels of inflammatory cytokines showed a downward trend, among which the high dose of SSP could significantly inhibit the secretion levels of inflammatory cytokines, IL-6, IL-1 β and TNF- α ($p < 0.01$), the above results indicated that SSP can significantly inhibit the inflammatory response of BMDCs *in vitro*.

Regulation of E-cadherin by SSP on DCs is influenced by miR-505-3p

After BMDCs stimulated by LPS, the relative expression level of miR-505-3p increased remarkably, which was consistent with the previous transcriptomic sequencing results and the trend of the relative expression level of miR-505-3p in mouse colon tissues (Fig. 9A). After SSP treatment, the relative expression of miR-505-3p in the different doses of SSP was noticeably decreased (Fig. 9A). As the encoding gene of E-

cadherin protein, the relative expression of *CDH1* on DCs also showed a obvious increase after LPS stimulation, and the increase was reversed by SSP (Fig. 9B). To further verified whether such changes were also reflected at the protein level, the expression of E-cadherin on BMDCs was detected by WB. Compared with the Vehicle group, the expression of E-cadherin in the LPS group increased notably, whereas under the intervention of different doses of SSP, the level of E-cadherin all showed a decreasing trend and showed a certain dose-dependent relationship, with the high dose having the better inhibition effect on E-cadherin (Fig. 9C-D). Combined with the previous results, the high-expressed miR-505-3p, *CDH1* and E-cadherin are closely related to the inflammatory status of BMDCs. SSP can simultaneously inhibit the expression of miR-505-3p, *CDH1*, and E-cadherin, and the regulatory effects show a consistent trend in vivo and vitro experiments.

To further understand whether the expression level of miR-505-3p has a direct effect on the cellular function of BMDCs, the surface markers related to inflammatory differentiation on knockdown BMDCs were detected by flow cytometry. Compared with the Vehicle group, the expression levels of CD86, MHC-II, TNF- α , and IL-12 in the LPS group were increased, and the expression level of the anti-inflammatory cytokine IL-10 was noticeably decreased (Fig. 9E1-I2). BMDCs knocked down with miR-505-3p were still able to develop inflammatory state and release inflammatory factors after stimulation with LPS, which indicated that the low expression of miR-505-3p level could not directly affect the inflammatory differentiation of BMDCs. After SSP administration intervention, compared with the LPS group, the expression levels of CD86, MHC-II, TNF- α , IL-12 and IL-10 only showed a slight downward trend, and did not form a remarkable difference with the LPS group (Fig. 9E1-I2), which indicated that the low expression of miR-505-3p affected the effect of the intervention of SSP on the inflammatory

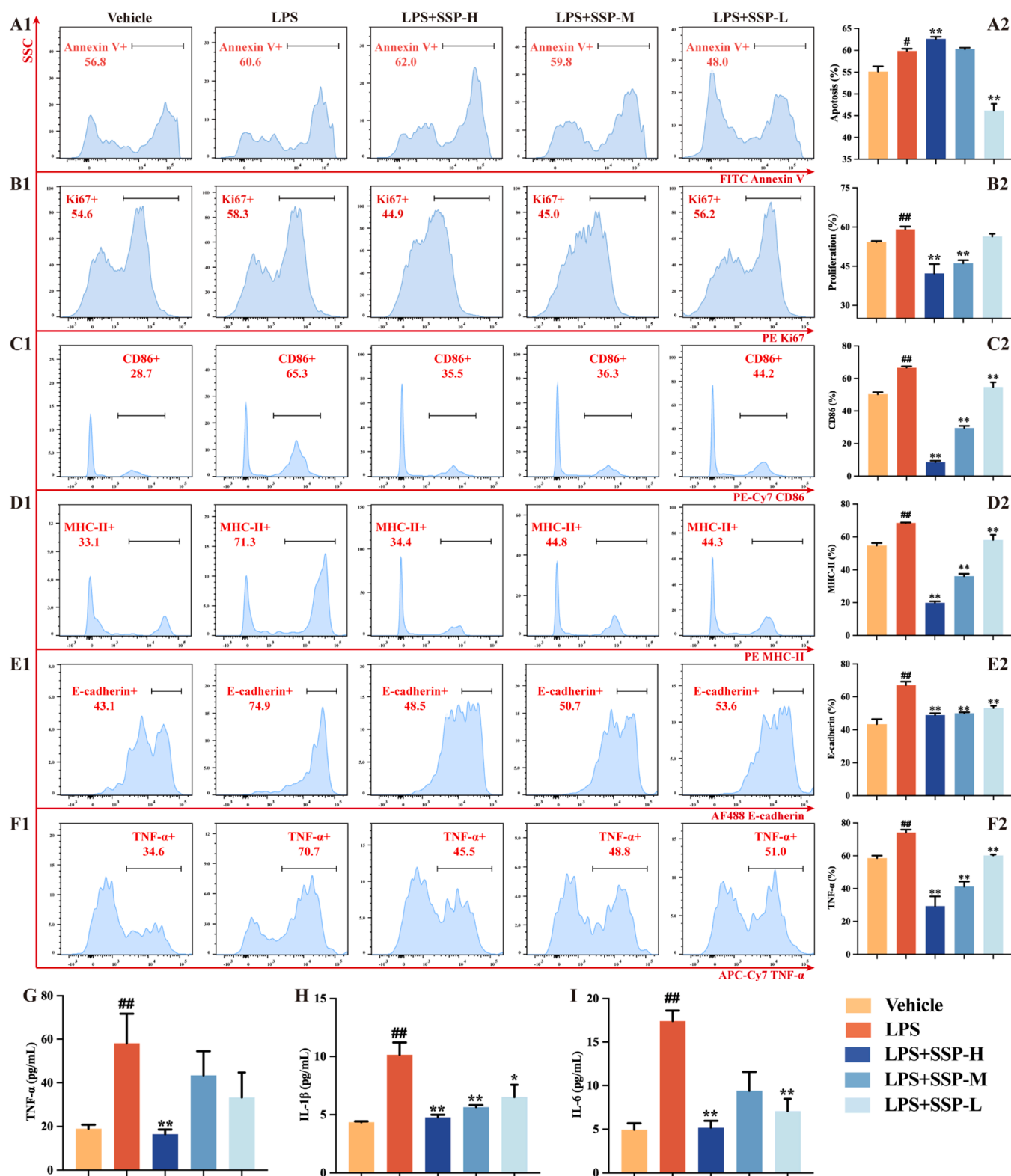


Fig. 8. Regulation of DCs maturation and inflammatory differentiation by SSP ($n = 3$). Representative flow cytometry peak map of Annexin V (A1), Ki67 (B1), CD86 (C1), MHC-II (D1), E-cadherin (E1), and TNF- α (E1) in each group. Flow cytometry histograms of Annexin V (A2), Ki67 (B2), CD86 (C2), MHC-II (D2), E-cadherin (E2), and TNF- α (E2) in each group. The expression levels of inflammatory cytokines TNF- α (G), IL-1 β (H), and IL-6 (I) of BMDCs in each group. Compared with Normal group, $^{\#}P < 0.05$, $^{##}P < 0.01$; Compared with DSS group, $^{*}P < 0.05$, $^{**}P < 0.01$.

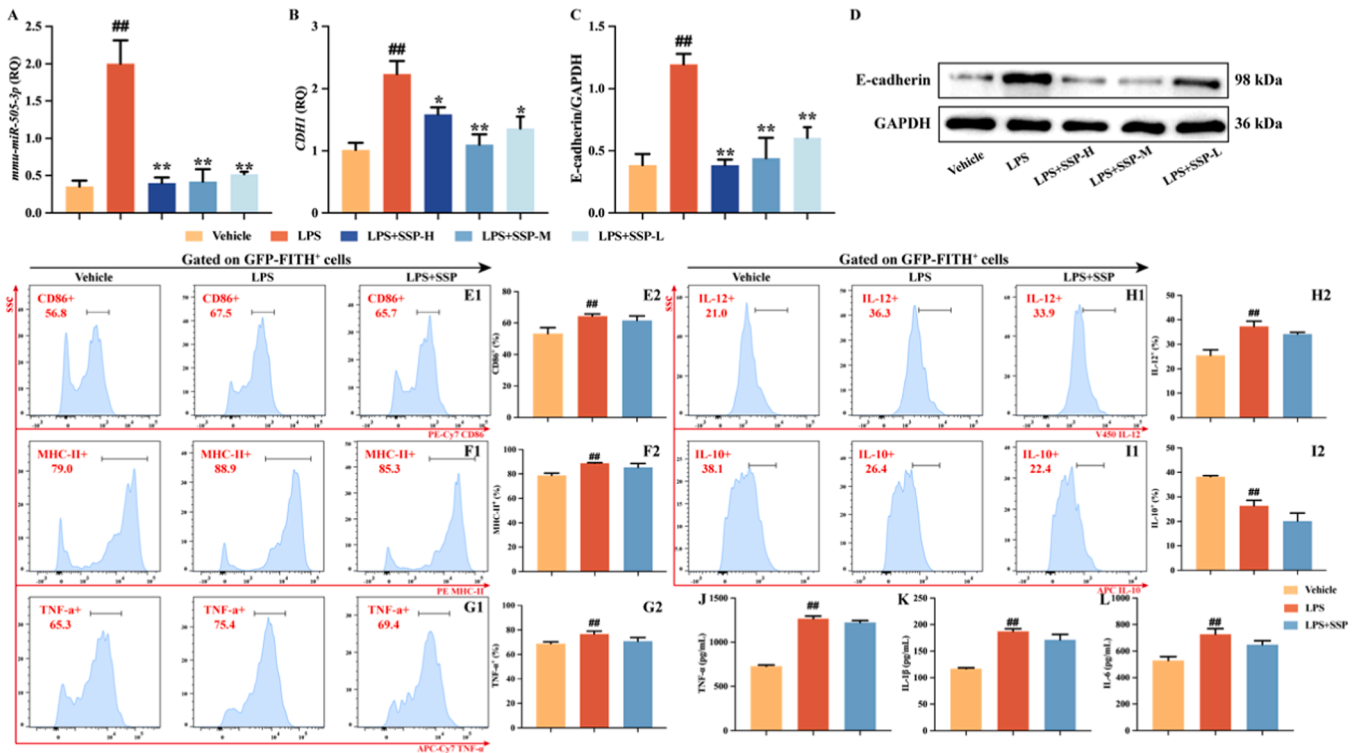


Fig. 9. Regulation of E-cadherin by SSP on DCs is influenced by *miR-505-3p* ($n = 3$). (A) Relative expression of *mmu-miR-505-3p* in BMDCs; (B) Relative expression of CDH1 in BMDCs; (C) Relative expression of E-cadherin protein in BMDCs; (D) Representative western blotting plot of E-cadherin expression in BMDCs. Representative flow cytometry peak map of CD86 (E1), MHC-II (F1), TNF- α (G1), IL-12 (H1), and IL-10 (I1) in each group. Flow cytometry histograms of CD86 (E2), MHC-II (F2), TNF- α (G2), IL-12 (H2), and IL-10 (I2); The expression levels of cytokines TNF- α (J), IL-1 β (K), and IL-6 (L) of BMDCs in each group. Compared with Normal group, $^*P < 0.05$, $^{**}P < 0.01$; Compared with DSS group, $^*P < 0.05$, $^{**}P < 0.01$.

differentiation of BMDCs, and that the SSP can no longer exert the corresponding regulatory effect at this time.

Meanwhile, compared with the Vehicle group, the levels of IL-6, IL-1 β and TNF- α in the LPS group showed an obvious increase; after treatment with SSP, the levels of IL-6, IL-1 β and TNF- α secreted by BMDCs in the LPS + SSP group were statistically different compared with those in the LPS group (Fig. 9J-L). At this point, SSP lost its inhibitory effect on the secretion level of inflammatory cytokines in BMDCs. The above experimental results demonstrated that SSP regulates the inflammatory differentiation status of DCs mainly by mainly regulating *miR-505-3p* to inhibit the expression of E-cadherin.

Validation of key active ingredients in SSP

Molecular docking methods were used to verify the affinity of the identified components of the SSP to the key targets. As shown in Fig. 10A, the top three strong binding capacity pairs of the identified components and each key target were as follows: Evodiamine (EVO) to mTOR (−10.4 kcal/mol), AKT1 (−9.7 kcal/mol) and PTPRC (−9.0 kcal/mol); Isobavachalcone to mTOR (−9.9 kcal/mol), AKT1 (−9.2 kcal/mol), MAPK3 (−8.4 kcal/mol) PPARC (−8.4 kcal/mol); Curcumin to AKT1 (−9.6 kcal/mol), MAPK3 (−9.3 kcal/mol), and mTOR (−9.2 kcal/mol). Eugenol to AKT1 (−6.6 kcal/mol), mTOR (−6.0 kcal/mol), ITGB1 (−6.0 kcal/mol) and JUN (−6.0 kcal/mol). In addition, the top three active ingredients that strongly bind to CDH1 are EVO (−7.4 kcal/mol), Isobavachalcone (−6.3 kcal/mol) and Curcumin (−6.9 kcal/mol). The above results indicated that EVO, Isobavachalcone and Curcumin played a dominant role in binding to the key targets above, and had a strong binding power to them, and EVO was the most prominent and combined best with CDH1. The representative molecular docking results for mTOR, AKT1, and CDH1 binding to the top two most stable active components were illustrated in Fig. 10B.

Subsequently, BMDC was intervened with EVO to verify whether it is the key component in SSP that inhibited E-cadherin by regulating *miR-505-3p*. In Fig. 9C-E, the relative expression of *miR-505-3p*, *CDH1*, and *TNF- α* was conspicuously increased in BMDC after LPS stimulation, which was well inhibited by both high and low doses of EVO. This is consistent with the regulatory effects of SSP on *miR-505-3p*, *CDH1* as well as *TNF- α* on BMDCs, so that EVO may be the key component of SSP in the treatment of UC.

Therapeutic effects of key active component of SSP in UC mice

As before, after DSS successfully induced colitis in mice, all treatment groups except the normal group exhibited weight loss, increased DAI score, splenomegaly accompanied by increased spleen weight, elevated spleen weight index, shortened colon length, and elevated colon weight index (Fig. 11A-H). Moreover, H&E Staining images showed that DSS caused extensive ulcerative damage to the colon, obvious inflammatory cell infiltration as well as thickening of the colon wall, and the pathological damage score was significantly increased (Fig. 11I-J). Compared with the DSS group, the administration of high, medium, and low doses of EVO had similar effects as 5-ASA-treated group, all of which markedly improved the various symptoms caused by DSS, such as body weight restoration, decrease in spleen weight index, restoration of colon length, and decrease in colon weight index (Fig. 11A-H). Meanwhile, H&E staining images showed that the EVO-treated groups at different doses exhibited less inflammatory cell infiltration, relatively intact colonic structure, less mucosal damage, and lower histologic scores (Fig. 11I and J). These results suggest that EVO effectively attenuated DSS-induced experimental colitis.

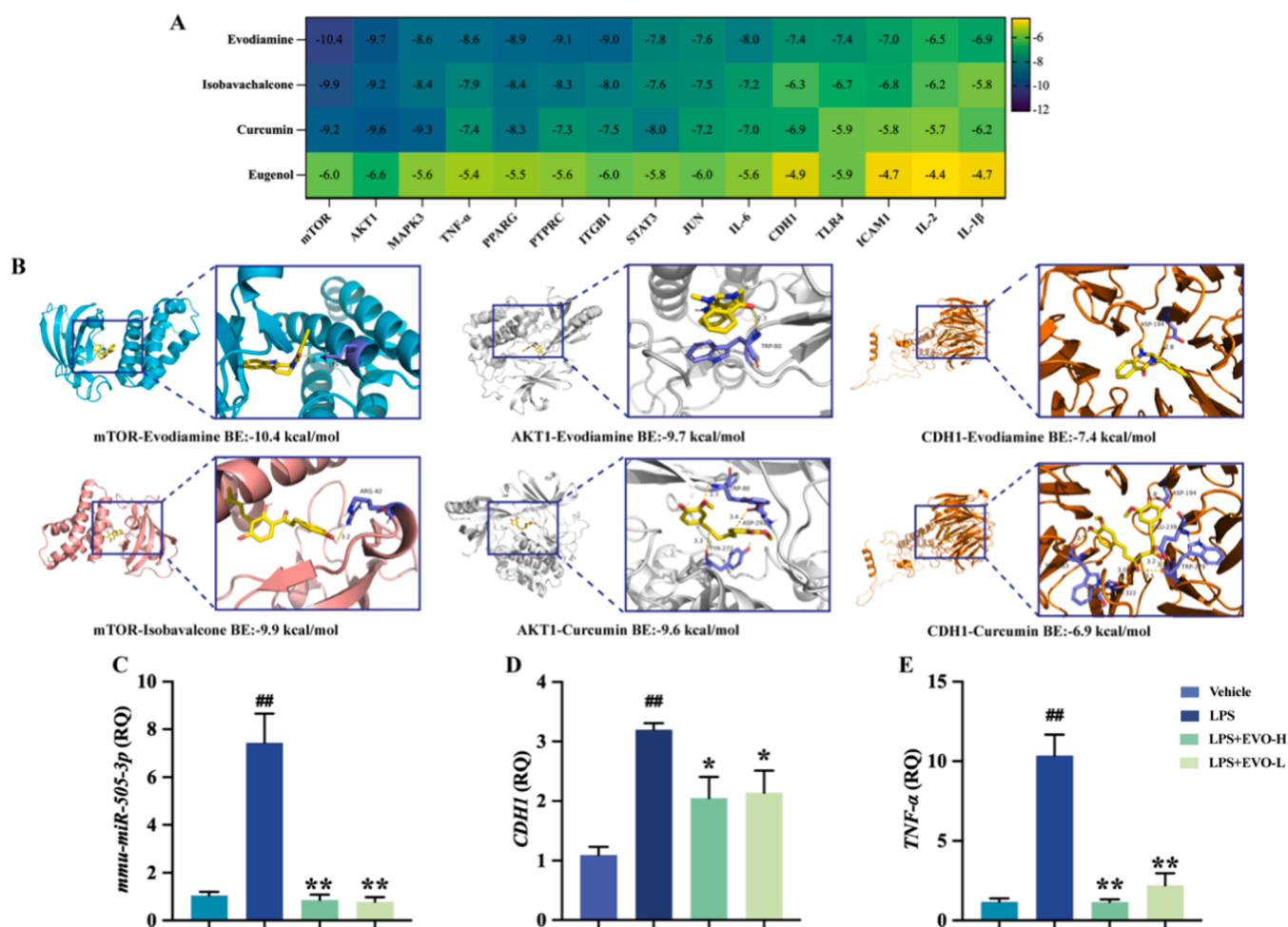


Fig. 10. Molecular docking results of key active ingredients and targets in SSP (A) The heatmap of docking scores of key targets combining to 4 active compounds in SSP. (B) The representative docking complex of key targets and compounds. (C) The relative expression of *mmu-miR-505-3p* on BMDCs in each group ($n = 3$). (D) The relative expression of *CDH1* on BMDCs in each group ($n = 3$). (E) The relative expression of *TNF-α* on BMDCs in each group ($n = 3$). Compared with Normal group, $\#P < 0.05$, $\#\#P < 0.01$; Compared with DSS group, $*P < 0.05$, $**P < 0.01$.

Regulation of key active ingredients on DC in UC mice

As previous, $CD11c^{+}CD103^{+}$ was used to gate DCs in lymph node mononuclear cells. In the DSS group, the cell frequencies of E-cadherin, MHC-II, FcγRI, CD24 and CD115 positive expression were obviously higher than those in the normal group (Fig. 12A1-E2). On the contrary, compared with the DSS group, different doses of EVO down-regulated the expression of E-cadherin, MHC-II, FcγRI, CD24 and CD115 (Fig. 12A1-E2), which was consistent with the regulatory trend of SSP on DCs. The above results demonstrated that EVO effectively inhibited the inflammatory DCs of mice with DSS-induced colitis, which may be a key active ingredient of SSP treated mice with UC by restraining *miR-505-3p* to inhibit E-cadherin⁺DCs.

Discussion

This study confirmed that SSP can effectively alleviate DSS induced UC in mice. SSP inhibited the expression of E-cadherin by down-regulating *miR-505-3p*, thereby preventing the inflammatory differentiation of DCs, reducing the secretion of pro-inflammatory cytokines, and alleviating intestinal inflammation. Meanwhile, through the analysis of transcriptomics and network pharmacology, the effective ingredients and key regulatory targets of SSP were screened, and then therapeutic effects of the key effective ingredient evodiamine on ulcerative colitis were verified in animal experiments. This study provided

solid scientific evidence for the future development and clinical application of SSP (Fig. 13).

UC is an inflammatory bowel disease with unknown pathogenesis, which is still difficult to cure and characterized by recurrent episodes. With the number of patients increasing year by year, UC has become a major challenge for the world's healthcare industry, increasing the burden on national healthcare economies (Parveen et al., 2020). Although aminosalicic acid, glucocorticoids, and various new immunosuppressive drugs have been used in the clinical treatment of UC, many patients still show different degrees of intolerable symptoms due to individual differences (Aggarwal et al., 2017; D'Amico et al., 2020; Zhou et al., 2021). SSP, an ancient classic prescription of Chinese medicine, is recognized as a preferred alternative drug for the treatment of UC in chronic remission, and it has a variety of pharmacological effects, including antioxidant, antibacterial, anti-inflammatory, and anti-diarrheal effects (Zhang et al., 2024a). In our previous study, it has been proved that SSP has a good immune-modulating function and affected on many immune cells (Chen et al., 2020; Ge et al., 2020). In this study, we explored the mechanism of SSP in the treatment of UC.

In the present study, oral administration of DSS induced severe colitis symptoms in mice, including weight loss, bloody diarrhea, ulcer formation, epithelial cell loss, and impaired barrier integrity, and rapidly elicited an inflammatory immune response. According to the pre-experiment, it was found that 2.5 mg/kg of SSP has a good anti-UC effect, after the same DSS induction, the mice treated with SSP showed

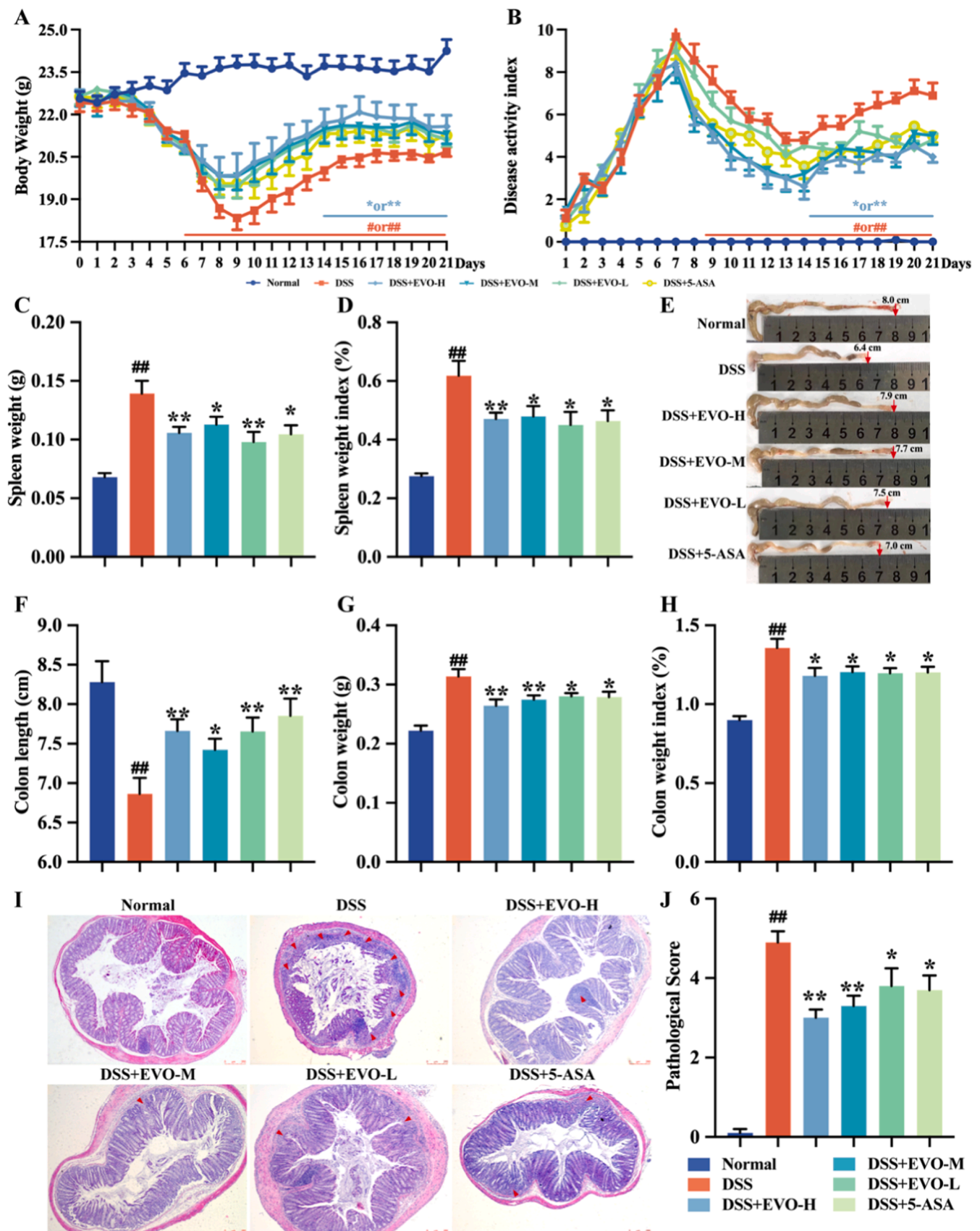


Fig. 11. Effects of EVO on DSS-induced UC ($n = 10$). (A) Body weight. (B) Disease activity score (DAI). (C) Spleen weight. (D) Spleen weight index. (E) Representative images of mice colon. (F) Colon length. (G) Colon weight. (H) Colon weight index. (I) Representative sections for hematoxylin and eosin (H&E) staining in each group. (J) Pathological score of colon tissue in each group. Compared with Normal group, * $P < 0.05$, ** $P < 0.01$; Compared with DSS group, * $P < 0.05$, ** $P < 0.01$.

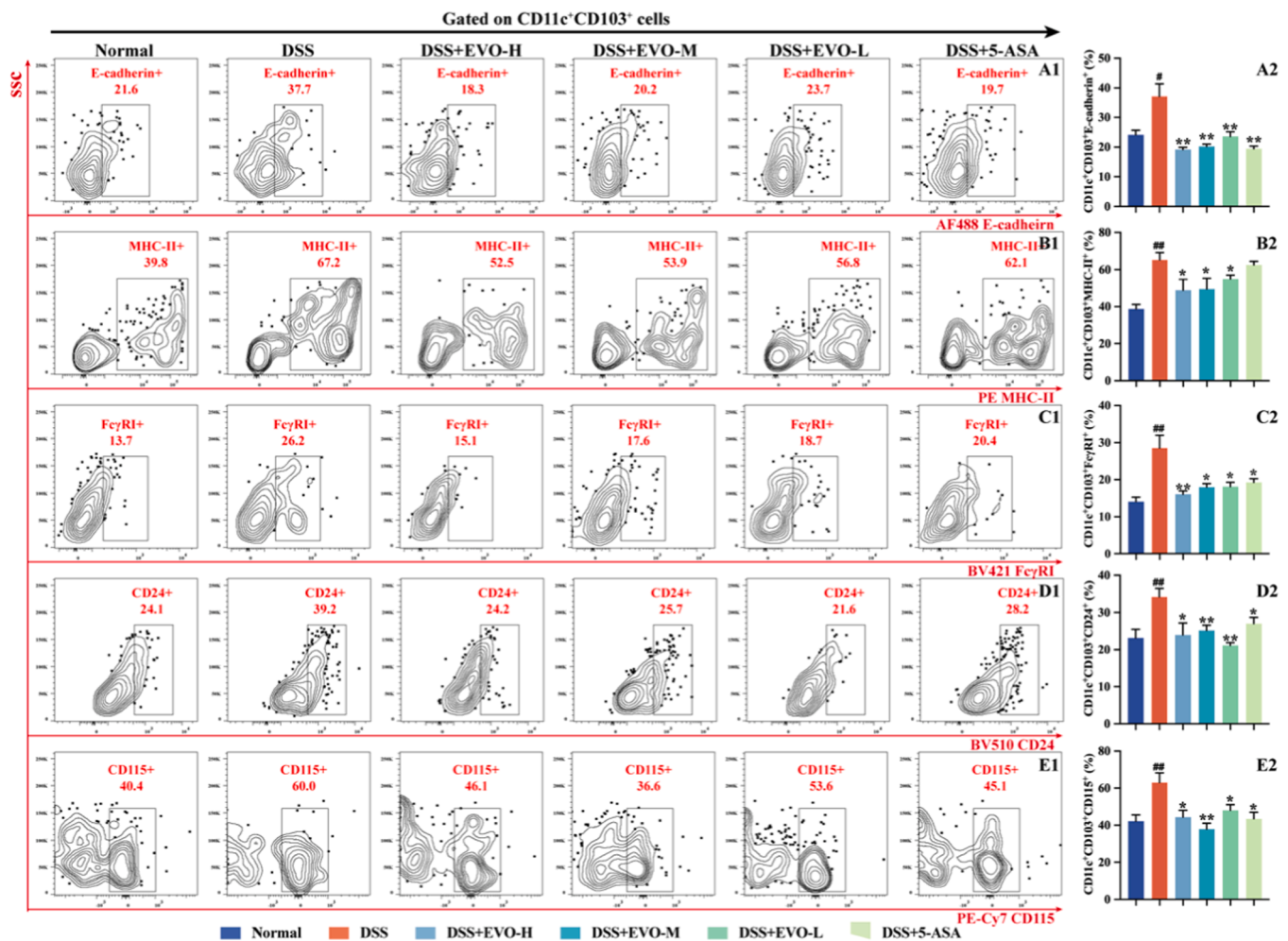


Fig. 12. Regulation of EVO on DCs in DSS induced UC mice ($n = 10$). Representative flow cytometry scatterplot of E-cadherin (A1), MHC-II (B1), FcγRI (C1), CD24 (D1) and CD115 (E1) in each group. Flow cytometry histograms of E-cadherin (A2), MHC-II (B2), FcγRI (C2), CD24 (D2) and CD115 (E2) in each group. Compared with Normal group, $^{\#}P < 0.05$, $^{##}P < 0.01$; Compared with DSS group, $^{*}P < 0.05$, $^{**}P < 0.01$.

significantly better symptom recovery than that of the DSS group, with significant recovery of body weight, lower OB scores, reduced colonic edema visible after dissection, and indicators such as colonic length and intestinal weight index recovered, and microscopically, the extent of ulceration of their colonic tissues was reduced, with a decrease in the infiltration of inflammatory cells, and the emergence of new crypts, indicating that the colon tissue is in a state of repair. Moreover, the inflammatory cytokines IL-6, IL-1 β and TNF- α showed a significant downward trend, indicating that SSP could effectively inhibit the secretion of inflammatory cytokines in colon cells of mice. SSP can effectively alleviate symptoms of UC without significant toxic side effects.

The gut is a natural immune defense line of the human body, involving various immune cells, and DCs are the immune cells with the strongest antigen-presenting function in the body. They can activate T cells to initiate immune responses and play a decisive role in the type and intensity of immune responses (Merad et al., 2013). When the immune function of DCs is dysregulated, it can trigger a series of immune overreactions and is one of the important factors in the pathogenesis of ulcerative colitis (Banchereau and Steinman, 1998). Studies have shown that CD11c⁺ DCs in the colon lamina propria of IBD patients express high levels of co-stimulatory molecules on the surface, and the cells are mature and activated, and can secrete large amounts of pro-inflammatory factors IL-6 and IL-12, thereby aggravating the intestinal inflammation of patients (Rescigno and Di Sabatino, 2009).

Inflammatory differentiation is an adaptive response of DCs when encountering inflammatory signals, which is crucial for the effective initiation and regulation of immune responses against infection or inflammation. It enhances the migratory ability of DCs to accumulate in large numbers toward the site of inflammation, as well as enhances the antigen-presenting function, which enables better activation of T cells and the release of a large number of inflammatory cytokines. It has been demonstrated that DCs in the inflammatory state highly express surface co-stimulatory molecules such as F4/80, SIRP α , FcγRI, FcεRI- α , Ly6c, and MHC-II, which are differentiated from monocytes at the site of inflammation only in the inflammatory environment and accumulate in large numbers in lymph nodes and inflamed tissues, and are able to mediate effector T cell differentiation (Cheng et al., 2023; Hespel and Moser, 2012; Krajina et al., 2003). Notably, E-cadherin⁺ DC, a typical inflammatory dendritic cell subpopulation that has attracted much attention in recent years, is able to produce large amounts of inflammatory cytokines, such as IL-6 and IL-23, through activation of relevant signaling pathways, and is also capable of inducing an immune response in intestinal Th17 cells, thereby exacerbating intestinal inflammation in UC patients. Meanwhile, E-cadherin, as a common cell adhesion molecule, regulates Langerhans cells maturation and migration, and its attachment to cells prevents BMDCs from inducing a tolerance state (Van den Bossche et al., 2012). Therefore, blocking the inflammatory differentiation of dendritic cells and inhibiting the proliferation and activation of inflammatory dendritic cells may be able to effectively

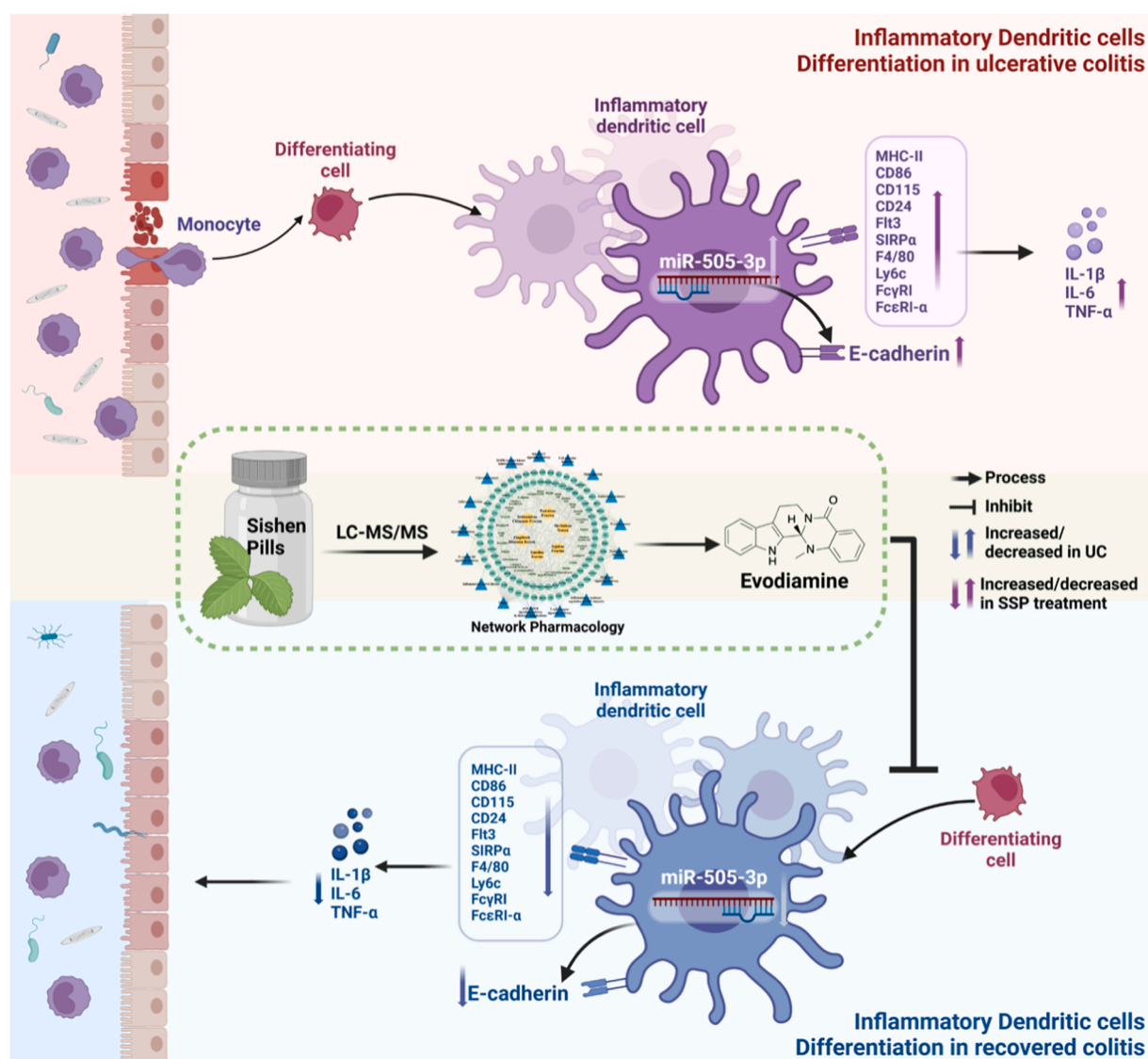


Fig. 13. The regulatory mechanism of SSP treatment for UC.

attenuate the inflammatory response, thus becoming a potential strategy for the treatment of UC.

To this end, we detected the surface co-stimulatory molecules associated with inflammatory differentiation of DCs in UC mice *in vivo*. The results showed that after DSS induction, the proportion of E-cadherin⁺ DC increased obviously compared with healthy mice. Meanwhile, the expression levels of SIRP-α, F4/80, Ly6c, MHC-II, FcγRI and FcεRI-α in DCs increased significantly. This suggests that DSS induced intestinal inflammation is directly related to the increase in the number of inflammatory DCs and the inflammatory differentiation of DCs. After SSP treatment, the number growth trend of inflammatory DCs was effectively reversed, and the expressions of SIRP-α, F4/80, Ly6c, MHC-II, FcγRI and FcεRI-α were noticeably decreased. Moreover, the expression of DCs surface markers related to maturation and inflammatory differentiation, such as CD115, CD24 and FcγRI, was significantly increased in the DSS group, while the expression was also effectively inhibited by SSP, indicating that SSP could effectively inhibit the inflammatory proliferation and differentiation of DCs. Furthermore, *in vitro* experiments, after LPS stimulation, BMDCs showed significant inflammatory proliferation and the expression of surface markers that label DCs maturation and inflammatory differentiation, such as CD86, MHC-II, E-cadherin, and TNF-α, were significantly elevated, whereas SSP showed a

significant inhibitory effect. Similarly, SSP significantly down-regulated the expression levels of inflammatory cytokines IL-6, IL-1β, and TNF-α in BMDCs. The results of both *in vivo* and *in vitro* experiments indicated that SSP was first found to treat UC by regulating the inflammatory differentiation of DCs.

miR-505-3p is part of miR-505 located on the X chromosome. Previous studies have shown that *miR-505* played a key role in regulating cell metabolism, growth, proliferation, and survival, and was involved in the development of a variety of diseases (Karni et al., 2008; Verduci et al., 2010; Yamamoto et al., 2011). *miR-505-3p* was closely related to tumor and cancers development, which was reported to be aberrantly expressed in breast cancer (Zhao et al., 2019), ovarian cancer (Xie et al., 2022), prostate cancer (Ninomiya et al., 2013), and glioblastoma (Mu et al., 2020). Studies on the biological function of *miR-505-3p* in UC were relatively rare, but there were still some genetic screening profiles of patients with IBD or UC showed that *miR-505-3p* was aberrantly expressed in UC. Compared to healthy control subjects, miR-505 activity was significantly lower in the blood of patients with active and inactive UC, and differed significantly between active UC and active CD, which can be used as a biomarker to distinguish between them (Dalal and Kwon, 2010; Wu et al., 2011). However, there is still lack of studies reported on whether *miR-505-3p* is directly involved in the mechanistic

study of UC.

In this study, by whole transcriptome sequencing, *miR-505-3p* was found to play a core regulatory role in the mechanism of SSP treating UC. Sequencing results showed that its expression was significantly increased in the DSS group and decreased in the DSS+SSP group compared with the Normal group, and showed the same expression trend in mouse colon tissues. A co-analysis was performed by using network pharmacology to predict the possible mechanism of SSP for UC. Interestingly, the candidate differential mRNAs highly overlapped with the possible targets predicted by network pharmacology, and KEGG and GO enrichment analyses demonstrated that they were closely related to DCs (such as PI3K-Akt signaling pathways, Jak-STAT signaling pathways, Rap1 signaling pathways), E-cadherin (such as Adherens junction, Tight junction), and inflammatory immune response (such as Acute inflammatory response, Regulation of inflammatory response, Response to bacterium). In addition, by searching from hTFtarget, transmiR and miRDB databases, it was found that there were a large number of shared transcription factors between *miR-505-3p* and *CDH1*, the gene encoding the E-cadherin protein, and most of them were involved in regulation of the E-cadherin. Therefore, combined with the results of network pharmacology and transcriptomics, we hypothesized that SSP alleviate UC by affecting the inflammatory differentiation of DCs, and which was closely related to E-cadherin. Furthermore, to explore the effect of the relationship between *miR-505-3p* and E-cadherin on the inflammatory differentiation of BMDCs, *miR-505-3p* expression was knocked down in BMDCs by lentiviral transfection in vitro experiments. Before transfection, the expression of *miR-505-3p*, *CDH1*, and E-cadherin were detected. After LPS stimulation, the expression levels of all three increased, while SSP could down-regulated their expression, and the regulation trend was consistent. After knocking down *miR-505-3p*, the levels of CD86, MHC-II, TNF- α , IL-12, and IL-10 on BMDCs were still increased after stimulation by LPS, while did not decrease in SSP-treated group. This indicted that knock-down of *miR-505-3p* did not affect the function of BMDCs in response to inflammation, but it prevented SSP from exerting an inhibitory effect on BMDC inflammatory differentiation. These results above demonstrated that the inhibitory effect of SSP on the inflammatory differentiation of BMDCs was achieved by down-regulating the expression of *miR-505-3p* to inhibit E-cadherin.

In addition, the active ingredients of SSP also showed potential for the treatment of UC in this study. Various active ingredients of SSP were identified by LC-MS/MS, and topological calculations of network pharmacology showed that among them, Evodiamine (EVO), Isobavachalcone, Curcumin and Eugenol could be the potential active ingredients in SSP. Assessing the binding capacity of active ingredients to key targets through molecular docking, the results indicated that EVO binds well to most of the key targets and binds most tightly to *CDH1*. In LPS-induced inflammatory cell model, the treatment of EVO conspicuously down-regulated the expression of *miR-505-3p*, *CDH1*, and *TNF- α* , which was consistent with the trend of SSP regulation on BMDCs. Similarly, in DSS-induced UC mice, different doses of EVO had obvious alleviating effects on inflammation in UC, and inhibited the inflammatory expression of DCs in vivo in UC mice, which was consistent with the regulatory effect of SSP. EVO was verified to be the key active components of SSP in regulating the inflammatory differentiation of DCs for UC. EVO showed favorable anti-UC pharmacological effects, which have been similarly reported in other studies. For example, EVO was able to treat UC by increasing the production of *Lactobacillus acidophilus* and protective acetate, regulating NF- κ B and NLRP3 inflammasome to inhibit inflammation (Wang et al., 2021). In this study, EVO was first found to be able to treat UC by modulating inflammatory differentiation of DCs through *miR-505-3p*, providing evidence for EVO to be a promising drug candidate of UC.

As a classic traditional Chinese medicine formula, SSP has a long history and is widely used as an alternative therapy in the clinical treatment of UC. However, research on its pharmacological mechanism is still very limited, greatly restricts the future development and

application of SSP. The novelty of this study lies in its research method and the discovery of potential targets and drug components: through the combination of network pharmacology and transcriptomics, the key regulatory target *miR-505-3p* and the active ingredient EVO were screened, and its regulatory mechanism was verified from different levels through animal experiments and cell experiments. In addition, E-cadherin⁺DCs, as a novel inflammatory DCs, provide a new direction for the study of the mechanism of SSP in treating UC.

However, there are still limitations in this study, such as the effect of E-cadherin expression on other functions of DCs and whether the regulatory effect of SSP on DCs is limited to inflammatory differentiation only. In future studies, we will define the steady-state effective constituents by LC-MS/MS, and used SPR, CO-IP, CETSA analyse to find the material basis inhibited the *miRNA-505-3p*-mediated E-cadherin to effectively treat UC.

Conclusion

In conclusion, SSP effectively alleviated DSS-induced colitis by inhibiting the inflammatory differentiation of DCs, which was potentially realized by restraining *miR-505-3p* to down-regulate E-cadherin on DCs, and one of the active components of SSP most likely is Evodiamine. Our findings provided evidence to clarify the mechanism of SSP for UC treatment and preliminarily verified the regulatory role of *miR-505-3p* and E-cadherin expression on DC cell function. In addition, potential active component (EVO) of SSP were identified and validated by network pharmacology and molecular docking, which provides a basis for the future development of EVO as a candidate natural anti-UC drug.

Supplementary materials: Supplementary material associated with this article can be found, in the online version, at doi:996 10.1016/j.phymed.2024.xxxxxx.

Data availability

The data included in the results of this study are available from the corresponding author upon request.

CRediT authorship contribution statement

Jiaqi Huang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation. **Youbao Zhong:** Writing – review & editing, Project administration, Investigation, Conceptualization. **Nian Cheng:** Project administration, Investigation, Formal analysis, Data curation. **Zheyang Zhang:** Project administration, Methodology, Investigation, Formal analysis. **Li Huang:** Methodology, Investigation, Formal analysis, Data curation. **Lizhao Song:** Investigation, Formal analysis, Data curation. **Shaomin Cheng:** Supervision, Resources, Project administration, Funding acquisition. **Haimei Zhao:** Supervision, Resources, Project administration, Funding acquisition. **Duanrong Liu:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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